

AI Meets Control Theory

From Classical Control to Intelligent Autonomous Systems

Living Technical Report — M15 — Dynamic Vehicle Handling & Rotary Inverted Pendulum

Generated by the project team

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September 5, 2026

Abstract

This report documents the culmination of the **AI Meets Control Theory** project. Across 10 curriculum modules, 28 rigorous from-scratch benchmark experiments, and 418 verified unit tests, we evaluate and compare classical control, modern state-space methods, Kalman filtering, optimal control, constrained Model Predictive Control (MPC), real-time iteration nonlinear MPC (iLQR / RTI-NMPC), adaptive control, Euler–Lagrange robot manipulators, non-holonomic mobile robotics, underactuated rotary inverted pendulums, high-speed dynamic vehicle handling with tire saturation, deep reinforcement learning, and hybrid AI + control safety shields on physical dynamical systems, under the core principle: *build every method from scratch, simulate it, visualise it, and compare it honestly.*

Contents

1	Introduction and Scope	2
2	Framework Architecture	3
2.1	Package Layout	3
3	Current Capabilities	4
4	Theory: Modelling and Simulation	6
5	Theory: Classical Control — PID Architecture	6
5.1	Worked Example: Stabilising an Unstable Plant (Experiment 03)	6
6	Theory: Modern Control, Estimation, and Optimal Design	7
6.1	Pole Placement and LQR	7
6.2	Worked Example: Cart-Pole Regulation (Experiment 04)	8
6.3	State Estimation: Observers, Kalman Filtering, and Duality	8
6.4	Worked Example: LQG on the Cart-Pole (Experiment 06)	9
6.5	Nonlinear Basin of Attraction: Cart-Pole LQR Robustness (Experiment 05) . . .	9
6.6	Nonlinear Control & Hybrid Swing-Up: Spong Energy Shaping (Experiment 07)	10
6.7	System Identification & Data-Driven Control (Experiment 09)	11
6.8	Constrained Model Predictive Control (Experiment 08)	11
7	Theory: Machine Learning and Learned Dynamics (Module 06)	12
7.1	Worked Example: Planning with Learned Dynamics vs. True Model (Experiment 10)	12

8 Theory: Reinforcement Learning for Dynamical Systems (Module 07)	13
8.1 Worked Example: Tabular Q-Learning vs. Classical Control on Pendulum (Experiment 11)	13
8.2 Worked Example: The RL Zoo vs. LQR on Cart-Pole Balance (Experiment 18)	14
9 Theory: AI + Control Hybrid Architectures (Module 08)	15
9.1 Worked Example: Shielded Tabular Q-Learning on Pendulum (Experiment 12)	15
10 Theory: Advanced State Estimation, Adaptive Control, and Robotics Capstones (Module 09)	16
10.1 Worked Example: Crazyflie 2.0 Flying a Lemniscate (Experiment 14)	16
10.2 Worked Example: EKF Output Feedback under Noisy Partial Sensing (Experiment 15)	17
10.3 Worked Example: EKF vs. UKF on Nonlinear Pendulum Estimation (Experiment 16)	18
10.4 Worked Example: Adaptive vs. Fixed Control on a Drifting Plant (Experiment 17)	19
10.5 Worked Example: Obstacle-Aware Nonlinear MPC on the Quadrotor (Experiment 20)	20
10.6 Worked Example: The Grand Capstone Bake-Off (Experiment 21)	21
11 Theory: The Intelligent Control Challenge (ICC) Benchmark (Module 10)	21
11.1 Worked Example: Multi-Paradigm Challenge Leaderboard (Experiment 19)	21
12 Phase 2: Real Multi-Body Systems and Method Depth	22
12.1 Worked Example: Differential-Drive Robot Path Following (Experiment 22)	22
12.2 Worked Example: Two-Link Planar Arm Tracking & Payload Adaptation (Experiment 23)	23
12.3 Worked Example: iLQR / Real-Time Iteration NMPC vs. Sampling MPC (Experiment 24)	23
12.4 Worked Example: iLQR vs. Sampling MPC on Harder Reference Paths (Experiment 26)	24
12.5 Worked Example: Moving-Obstacle Differential-Drive Avoidance (Experiment 25)	25
12.6 Worked Example: Dynamic Bicycle Vehicle Double Lane Change (Experiment 27)	26
12.7 Worked Example: Furuta Rotary Inverted Pendulum Benchmark (Experiment 28)	28
13 Engineering Synthesis & Decision Guide	29
13.1 Recurring Engineering Lessons Across 28 Experiments	30
14 Project Status and Roadmap	31

1 Introduction and Scope

The central question of the project is: **when should we use classical control, when should we use AI, and when should we use both?** Rather than assuming a neural network or an RL agent is the answer, every problem is first attacked with PID, state feedback, observers, LQR, Kalman filtering (linear, EKF, UKF), system identification, adaptive control (MRAC, Slotine–Li), nonlinear energy shaping, computed torque, constrained linear MPC, real-time iteration nonlinear MPC (iLQR / RTI-NMPC), sampling-based nonlinear MPC with obstacle avoidance, learned neural dynamics, deep reinforcement learning (DQN, PPO), behavioral cloning (BC),

and safety shields, and each method is measured against the others under identical initial conditions, sensor suites, noise profiles, disturbances, and actuator constraints.

Every topic follows the standardized engineering cycle:

THEORY $\rightarrow$ **DERIVATION** $\rightarrow$ **IMPLEMENTATION** $\rightarrow$ **SIMULATION**
 $\rightarrow$ **VISUALISATION** $\rightarrow$ **VALIDATION** $\rightarrow$ **COMPARISON** $\rightarrow$ **EXPERIMENT**

Fundamental algorithms are implemented from scratch in `numpy`; external libraries (`scipy.linalg`, `scipy.signal`, `control`, `gymnasium`, `stable-baselines3`) are used strictly for verification and benchmarking compliance.

2 Framework Architecture

The reusable library is the Python package `aimct` (`src/aimct/`). Its architecture separates distinct concerns behind minimal, strongly-typed interfaces so that dynamical systems, controllers, state estimators, neural network models, reinforcement learning environments, safety shields, and the simulator compose seamlessly.

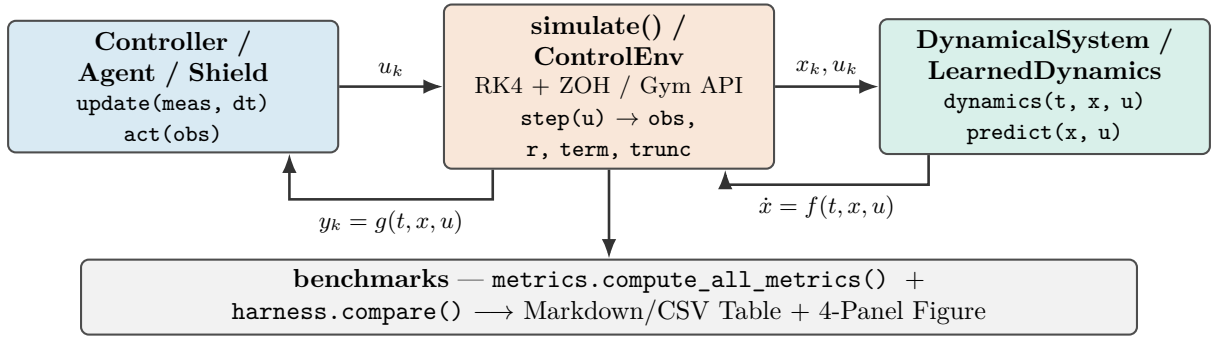


Figure 1: Core framework architecture. The numerical simulation engine drives a controller (`update()`) and continuous plant or learned neural surrogate (`dynamics()`) via fixed-step 4th-order Runge–Kutta with Zero-Order Hold (ZOH).

2.1 Package Layout

```

src/aimct/
  systems/      DynamicalSystem base + LinearSystem, MassSpringDamper,
                 Pendulum, CartPole, PlanarQuadrotor (Crazyflie 2.0), DCMotor,
                 DifferentialDriveRobot (unicycle + lag), TwoLinkArm (Euler-Lagrange)
  simulate.py   rk4_step(), simulate() -> Trajectory(t, x, u, y)
  controllers/  PID, StateFeedback, LQR, ObserverFeedback, MRAC, ComputedTorque,
                 EnergyShapingSwingUp, HybridSwingUpLQR,
                 LinearMPC (with preview), SamplingMPC (CEM + obstacles),
                 ILQR (trajectory optimiser + real-time-iteration NMPC)
  estimation/   LuenbergerObserver, KalmanFilter (LQE/FARE), DiscreteKalmanFilter,
                 ExtendedKalmanFilter, UnscentedKalmanFilter, observability_matrix
  trajectories/ Lemniscate, Spline, Minimum-Jerk Polynomials, Dubins paths
  sysid/        least_squares_id, dmcdc, to_continuous (block logm), prediction_error
  ml/           MLP (backprop + Adam), LearnedDynamics (grey-box / residual)
  rl/           ControlEnv, Discretizer, QLearning, DQN, REINFORCE, PPO, BC
  hybrid/       ShieldedController (switch/filter blends, predicate helpers)
  viz/          SystemArtist contract (Pendulum, CartPole, TwoLinkArm, DiffDrive,
                 PlanarQuadrotor), animate() replay generator, Sandbox live GUI
  dev/          Design-time preview dashboard (poles, controllability, Jacobian
  residuals)

```

```

benchmarks/  metrics.py (13 metrics), harness.py, sweep.py, challenge.py (ICC
  engine),
              tracking.py (path following benchmark), capstone_scoring.py
plot_style.py Okabe-Ito palette + publication-ready 4-panel comparison figure

```

3 Current Capabilities

Table 1 summarises the implemented components, their mathematical scope, and the corresponding verification test coverage as of M15 — Dynamic Vehicle Handling & Rotary Inverted Pendulum.

Component	What it provides	Tests
<code>aimct.systems</code>	Continuous-time ODE models with a common <code>dynamics(t, x, u)</code> interface, full-state or output projections, and analytical/numerical <code>linearize()</code> . Ships: <code>LinearSystem</code> , <code>MassSpringDamper</code> , <code>Pendulum</code> , <code>CartPole</code> , <code>PlanarQuadrotor</code> (Crazyflie 2.0), <code>DCMotor</code> , <code>DifferentialDriveRobot</code> (unicycle + motor lag), <code>TwoLinkArm</code> (Euler–Lagrange with wrist payload).	✓
<code>aimct.simulate</code>	Fixed-step RK4 integrator with zero-order hold; <code>Trajectory</code> container; hard actuator saturation; measurement and disturbance injection hooks.	✓
<code>aimct.controllers.PID</code>	From scratch: filtered derivative-on-measurement ($D(s) = \frac{K_d s}{\tau_d s + 1}$), conditional-integration anti-windup clamping, setpoint weighting.	✓
<code>aimct.controllers.StateFeedback</code>	Arbitrary closed-loop pole placement via Ackermann’s formula (SISO) with feedforward steady-state gain scaling k_r .	✓
<code>aimct.controllers.LQR</code>	Infinite-horizon optimal regulator; Continuous Algebraic Riccati Equation (CARE) solved from scratch via Hamiltonian Schur decomposition, cross-checked with <code>scipy.linalg</code> .	✓
<code>aimct.controllers.MRAC</code>	Model Reference Adaptive Control: Lyapunov-derived parameter adaptation tracking a reference model under drifting or uncertain plant parameters.	✓
<code>aimct.controllers.iLQR</code>	Iterative Linear Quadratic Regulator (iLQR) trajectory optimiser with quadratic-cost expansion, backward Riccati pass, forward line search, and Real-Time-Iteration (RTI-NMPC) wrapper.	✓
<code>aimct.controllers.ObserverFeedback</code>	Unified output feedback combining any state estimator (Luenberger, linear KF, EKF, UKF) with static feedback gain $u = -K\hat{x}$.	✓
<code>aimct.controllers.SwingUp</code>	Nonlinear Spong energy shaping (<code>EnergyShapingSwingUp</code>) with partial feedback linearisation, plus finite-state hysteresis supervisor (<code>HybridSwingUpLQR</code>).	✓
<code>aimct.controllers.LinearMPC</code>	Constrained linear MPC via condensed receding-horizon QP, active-set solver (<code>_qp.solve_qp</code>), DARE terminal cost, soft state constraints, and trajectory preview.	✓
<code>aimct.controllers.SamplingMPC</code>	Sampling-based nonlinear MPC using the Cross-Entropy Method (CEM), vectorized batch trajectory rollouts, variance floor, CARE terminal weight, and online non-convex obstacle avoidance.	✓
<code>aimct.estimation</code>	State estimation suite: <code>observability_matrix</code> , <code>is_observable</code> , <code>place_observer</code> , <code>LuenbergerObserver</code> , <code>solve_fare</code> , <code>KalmanFilter</code> , <code>DiscreteKalmanFilter</code> , <code>ExtendedKalmanFilter</code> , and <code>UnscentedKalmanFilter</code> ($2n + 1$ scaled sigma points, Merwe parameters, zero Jacobian linearisation).	✓
<code>aimct.trajectories</code>	Trajectory generators: Lemniscate, minimum-jerk polynomials, Dubins paths, parametric splines.	✓
<code>aimct.sysid</code>	Data-driven system identification: <code>least_squares_id</code> , <code>dmdc</code> (rank-truncated SVD), <code>to_continuous</code> (exact block matrix logarithm), <code>model_mismatch</code> .	✓
<code>aimct.ml</code>	Machine learning suite: multi-layer perceptron (MLP) with hand-written backpropagation + Adam, and residual / grey-box forward dynamics predictor (<code>LearnedDynamics</code> with <code>base_step</code>).	✓
<code>aimct.rl</code>	Reinforcement learning suite: <code>Gymnasium ControlEnv</code> , <code>Discretizer</code> , <code>Tabular QLearning</code> , <code>DQN</code> (experience replay + target net), <code>REINFORCE</code> , <code>PPO</code> (clipped surrogate objective + GAE), and <code>Behavioral Cloning</code> (BC).	✓
<code>aimct.hybrid</code>	Hybrid AI + control safety shields: <code>ShieldedController</code> (switch and filter action blends, predicate helpers <code>box_predicate</code> , <code>barrier_predicate</code> , and audit logging).	✓

4 Theory: Modelling and Simulation

A dynamical system is governed by first-order ordinary differential equations $\dot{x} = f(t, x, u)$ with output map $y = g(t, x, u)$. Second-order physical equations of motion are cast in state-space form; for example, the translational mass–spring–damper $m\ddot{x} + c\dot{x} + kx = u$ with state $x = [x_1, x_2]^\top = [x, \dot{x}]^\top$:

$$\frac{d}{dt} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} u.$$

Jacobian Linearisation. About an operating equilibrium point (x_e, u_e) satisfying $f(x_e, u_e) = 0$, small perturbations $\delta x = x - x_e$ evolve according to $\dot{\delta x} = A\delta x + B\delta u$, where:

$$A \triangleq \left. \frac{\partial f}{\partial x} \right|_{(x_e, u_e)}, \quad B \triangleq \left. \frac{\partial f}{\partial u} \right|_{(x_e, u_e)}.$$

The `DynamicalSystem` base class evaluates A, B numerically via central differencing ($\epsilon = 10^{-6}$), while concrete models implement analytical Jacobians.

Numerical Integration. The continuous dynamics are integrated over time steps $h = \Delta t$ using explicit 4th-order Runge–Kutta (RK4):

$$k_1 = f(t_k, x_k, u_k), \quad k_2 = f\left(t_k + \frac{h}{2}, x_k + \frac{h}{2}k_1, u_k\right), \quad k_3 = f\left(t_k + \frac{h}{2}, x_k + \frac{h}{2}k_2, u_k\right), \quad k_4 = f(t_k + h, x_k + hk_3, u_k),$$

$$x_{k+1} = x_k + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

Holding control input $u(t) \equiv u_k$ constant across each interval $[t_k, t_k + h)$ reflects the physical Zero-Order Hold (ZOH) of digital microcontrollers.

5 Theory: Classical Control — PID Architecture

The continuous-time PID control law with filtered derivative is:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}, \quad e(t) = r(t) - y(t).$$

In discrete time with sampling interval h , derivative action is applied directly to the measured output with low-pass filter time constant $\tau_d = \frac{K_d}{N K_p}$ ($N = 10$) to prevent high-frequency noise amplification:

$$e_k = r_k - y_k, \quad D_k = \frac{\tau_d}{\tau_d + h} D_{k-1} + \frac{K_d}{\tau_d + h} (y_k - y_{k-1}), \quad u_{\text{unsat},k} = K_p e_k + I_k - D_k.$$

Anti-Windup Protection. When actuators reach physical limits ($|u| \leq u_{\text{max}}$), *conditional integration (clamping)* freezes the integrator update ($I_{k+1} = I_k$) whenever u_{unsat} is saturated and e_k has the same sign as u_k , preventing destructive overshoot upon setpoint arrival.

5.1 Worked Example: Stabilising an Unstable Plant (Experiment 03)

We evaluate the controller on an open-loop unstable plant with a Right-Half Plane (RHP) pole:

$$\ddot{y}(t) - \omega_0^2 y(t) = u(t) + d(t), \quad \omega_0 = 2.0 \text{ rad/s}, \quad G(s) = \frac{1}{s^2 - 4}.$$

The system is tested under a unit step reference $r = 1.0 \text{ rad}$, a step input disturbance torque $d = 0.5 \text{ N} \cdot \text{m}$ applied at $t = 5.0 \text{ s}$, and actuator saturation $|u(t)| \leq 10.0 \text{ N} \cdot \text{m}$.

Controller	Rise t_r [s]	Settle t_s [s]	Overshoot M_p	Steady e_{ss}	IAE	ITAE	Energy E_u	Sat. %
P-only ($K_p = 20$)	0.275	10.0	$9.54 \times 10^9\%$	6.04×10^7	4.77×10^7	4.53×10^8	984.3	97.62%
PD ($K_d = 5.66$)	0.389	10.0	28.19%	0.2812	2.811	13.63	316.3	1.60%
PID (no AW)	0.334	6.162	53.08%	3.90×10^{-5}	0.9274	1.043	262.2	5.55%
PID + Anti-Windup	0.362	6.161	39.43%	3.90×10^{-5}	0.7979	0.876	245.1	1.61%

Table 2: Benchmark results for Experiment 03 (regenerated from experiments/03_pid_stabilizes_unstable/table.md).

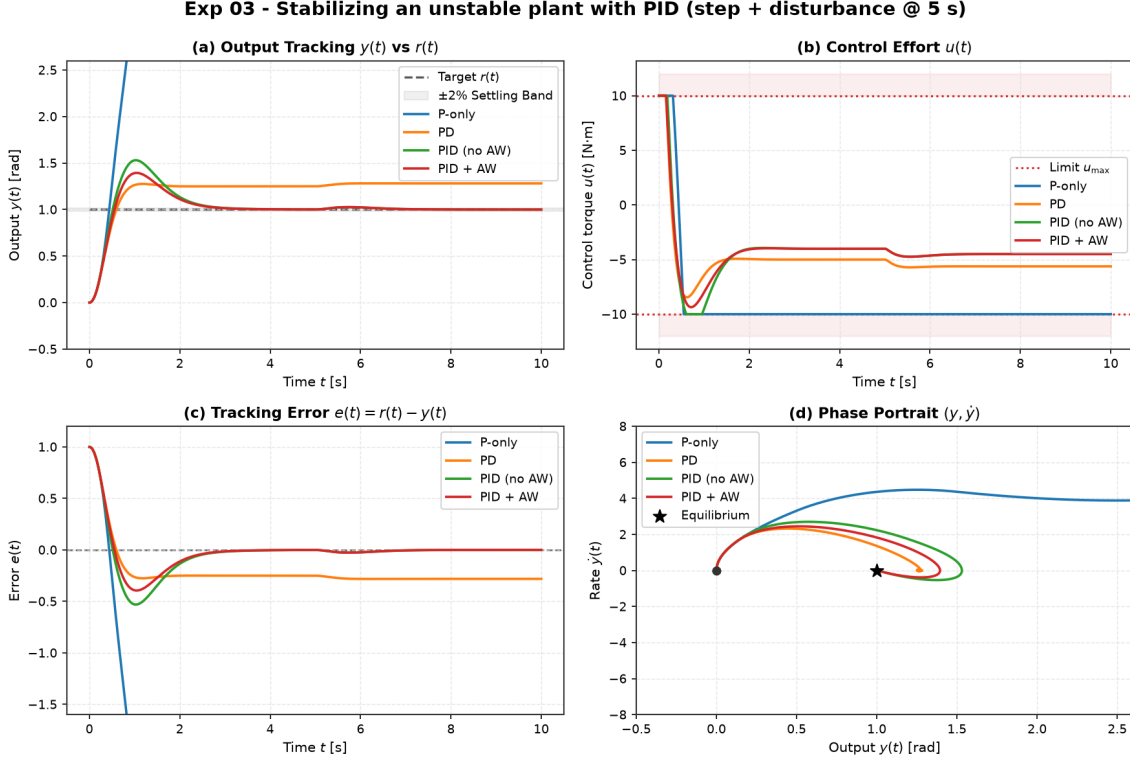


Figure 2: Experiment 03 benchmark comparison. (a) Output tracking $y(t)$, (b) Control effort $u(t)$ with saturation limits $\pm 10 \text{ N} \cdot \text{m}$, (c) Tracking error $e(t)$, and (d) Phase portrait $(y, \dot{y})$.

6 Theory: Modern Control, Estimation, and Optimal Design

6.1 Pole Placement and LQR

For controllable linear systems $\dot{x} = Ax + Bu$, full-state feedback $u = -Kx + k_r r$ assigns arbitrary closed-loop poles $\sigma(A - BK)$. **StateFeedback** computes K via Ackermann's formula:

$$K = \begin{bmatrix} 0 & \dots & 0 & 1 \end{bmatrix} \mathcal{C}(A, B)^{-1} \Delta_{\text{des}}(A), \quad k_r = -\frac{1}{C(A - BK)^{-1}B}.$$

Linear Quadratic Regulator (LQR). Rather than guessing pole locations, LQR computes the optimal feedback gain $K = R^{-1}B^\top P$ minimising the infinite-horizon quadratic cost:

$$J = \int_0^\infty \left(x(t)^\top Q x(t) + u(t)^\top R u(t) \right) dt, \quad Q \succeq 0, \quad R \succ 0.$$

The kernel $P = P^\top \succ 0$ uniquely solves the **Continuous Algebraic Riccati Equation (CARE)**:

$$A^\top P + PA - PBR^{-1}B^\top P + Q = 0.$$

We solve CARE from scratch via the stable invariant subspace of the Hamiltonian matrix:

$$\mathcal{H}_{\text{ham}} = \begin{bmatrix} A & -BR^{-1}B^\top \\ -Q & -A^\top \end{bmatrix} \in \mathbb{R}^{2n \times 2n}.$$

Continuous full-state LQR provides guaranteed robustness margins: gain margin $G_m \in [1/2, \infty)$ (-6 dB to $+\infty$ dB) and phase margin $\Phi_m \geq 60^\circ$ in all channels.

6.2 Worked Example: Cart-Pole Regulation (Experiment 04)

Controller	Settle θ t_s [s]	RMSE θ [rad]	Energy E_u [N^2s]	Peak $ u _{\max}$ [N]	Peak $ x _{\max}$ [m]	Status
Pole Placement	1.05	0.0171	2.34	8.71	0.162	Stable
LQR (Balanced)	1.05	0.0171	2.34	8.71	0.162	Stable
LQR (Soft)	1.70	0.0223	0.959	3.08	0.321	Stable
PID (Angle Only)	0.20	0.0110	8.76	18.00	0.823	Stable (Cart Drifts)

Table 3: Benchmark results for Experiment 04 on nonlinear Cart-Pole (from experiments/04_lqr_vs_pole_placement_cartpole/table.md).

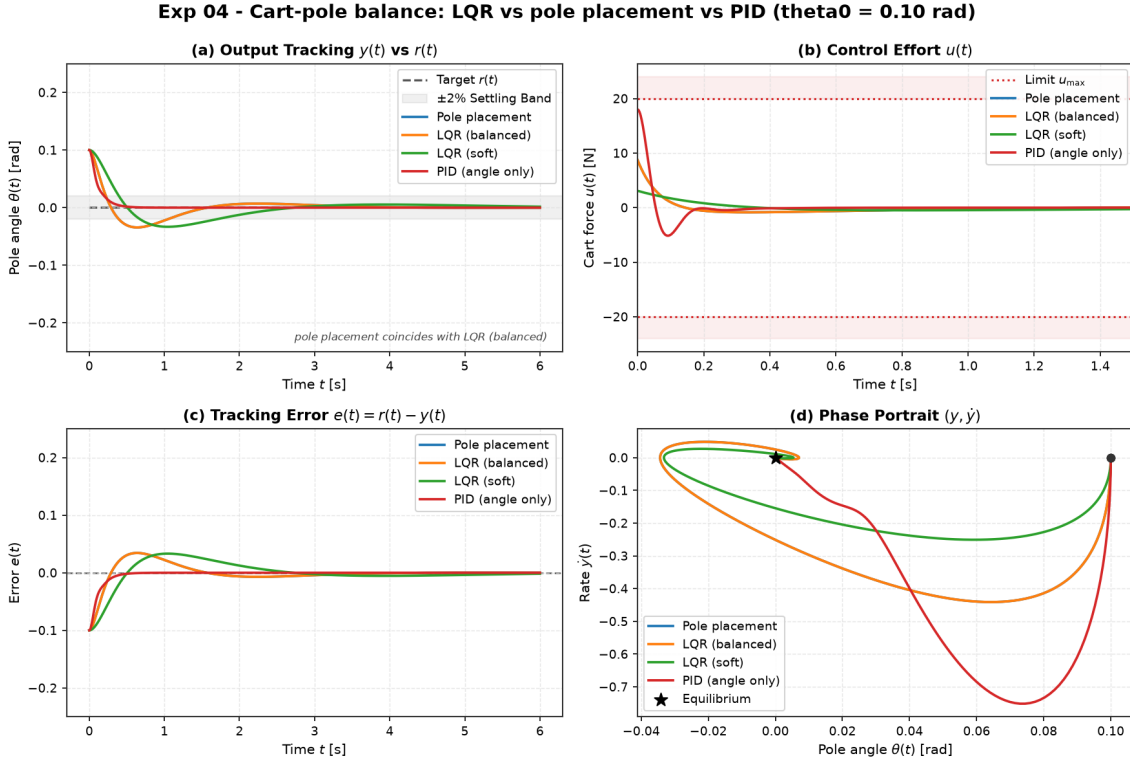


Figure 3: Experiment 04 benchmark comparison on nonlinear Cart-Pole.

6.3 State Estimation: Observers, Kalman Filtering, and Duality

Mathematical Duality: LQR Control vs. Kalman Filtering (LQE).

LQR Regulator (CARE): $A^\top P + PA - PBR^{-1}B^\top P + Q = 0, \quad K = R^{-1}B^\top P$

LQE Estimator (FARE): $AP + PA^\top - PC^\top R_v^{-1}CP + Q_w = 0, \quad L = PC^\top R_v^{-1}$

6.4 Worked Example: LQG on the Cart-Pole (Experiment 06)

Controller	Settle θ t_s [s]	RMSE θ [rad]	Energy E_u [N ² s]	Peak $ u _{\max}$ [N]	Status
LQR (Full State, Clean)	0.942	0.0150	1.49	6.97	Ideal Baseline
LQG (Optimal Filter)	2.070	0.0249	2.21	3.28	Optimal Smooth
Luenberger + K	0.496	0.0165	16.20	15.10	Severe Noise Chattering
LQG (Overconfident $V/100$)	0.828	0.0215	2.97	4.80	Noise Amplified

Table 4: Benchmark results for Experiment 06 comparing LQG, Luenberger observer, and clean LQR under measurement noise (from experiments/06_lqg_vs_lqr_measurement_noise/table.md).

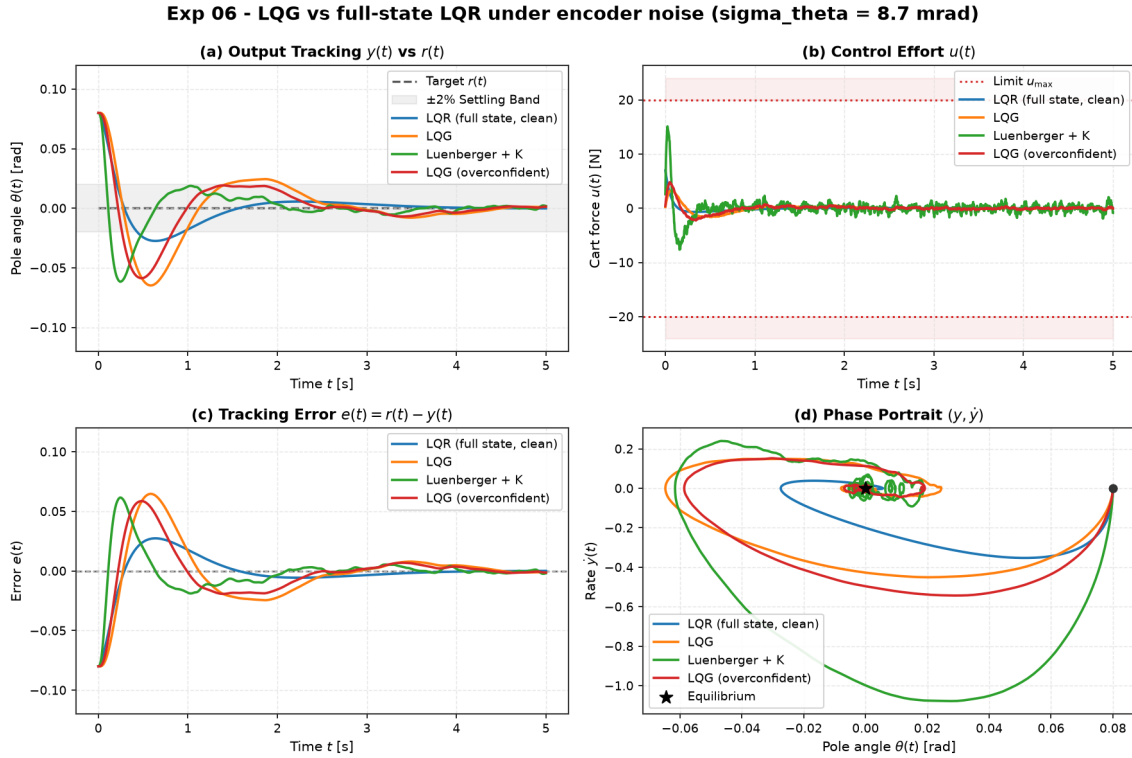
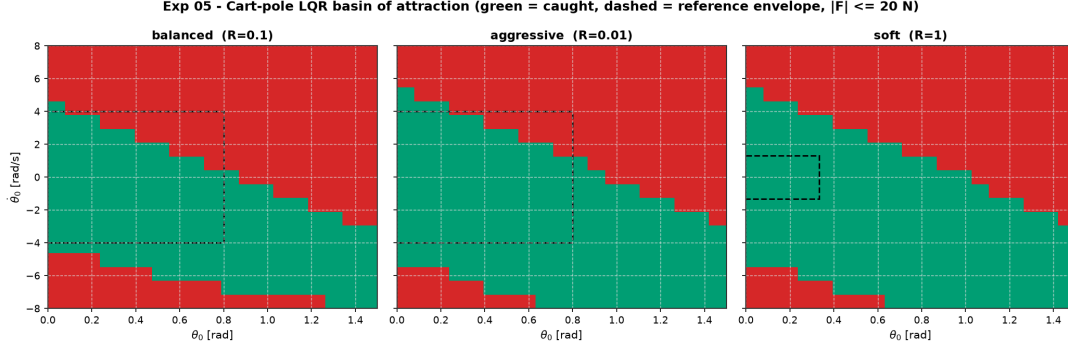


Figure 4: Experiment 06 benchmark comparison under encoder noise.

6.5 Nonlinear Basin of Attraction: Cart-Pole LQR Robustness (Experiment 05)

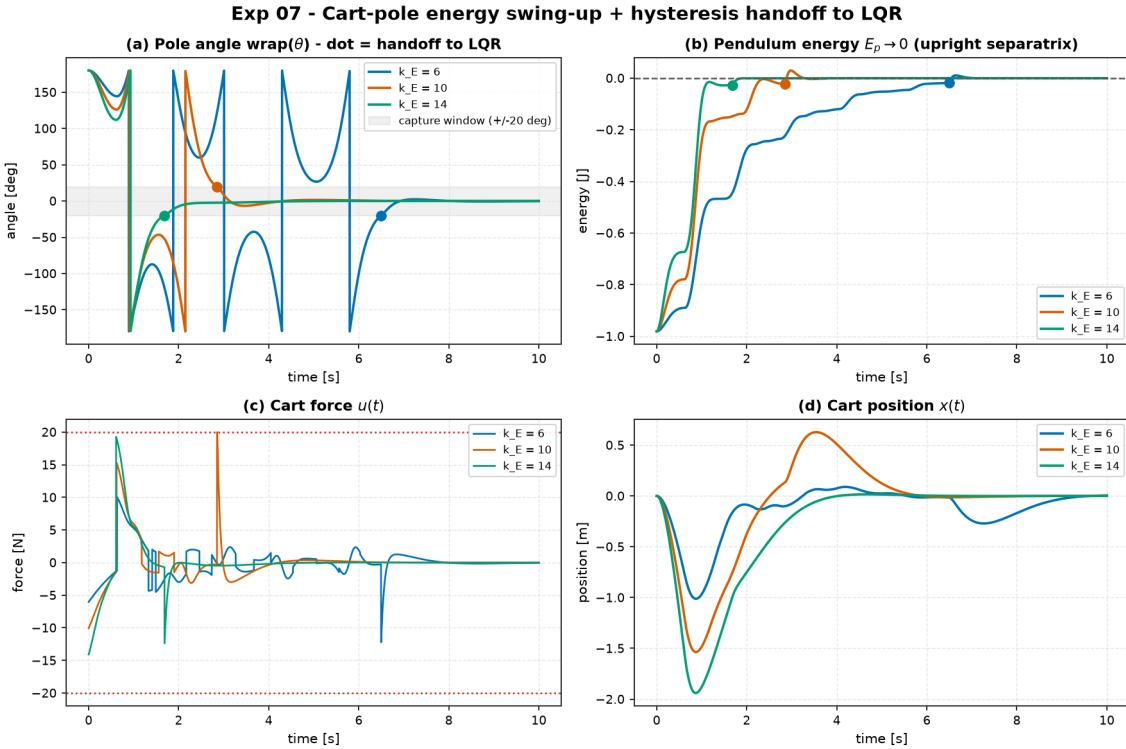
Tuning	Effort R	Measured θ_0 Edge	Lyapunov Est.	Measured $\dot{\theta}_0$ Edge	Lyapunov Est.
Balanced	0.10	0.83 rad (48°)	0.80 rad (46°)	4.4 rad/s	4.0 rad/s
Aggressive	0.01	0.92 rad (53°)	0.80 rad (46°)	5.3 rad/s	4.0 rad/s
Soft	1.00	1.00 rad (57°)	0.33 rad (19°)	5.3 rad/s	1.3 rad/s

Table 5: Measured recoverable boundaries vs. theoretical Lyapunov sub-level sets on nonlinear Cart-Pole.

Figure 5: Experiment 05 2D ($\theta_0 \times \dot{\theta}_0$) basin of attraction maps.

6.6 Nonlinear Control & Hybrid Swing-Up: Spong Energy Shaping (Experiment 07)

Energy Gain k_E	Capture t_{cap} [s]	Settle t_s [s]	Switches	Energy E_u [N^2s]	Peak $ u _{\text{max}}$ [N]	Cart Excursion [m]	Status
$k_E = 6$	6.496	7.452	1	52.28	12.23	1.012	Balanced
$k_E = 10$	2.852	4.058	1	87.28	20.00	1.536	Balanced
$k_E = 14$	1.686	3.344	1	107.24	19.27	1.940	Balanced

Table 6: Benchmark results for Experiment 07 swing-up from $\theta_0 = \pi$ across energy gains $k_E \in \{6, 10, 14\}$ under $|F| \leq 20$ N (from experiments/07_cartpole_swingup_hybrid/table.md).Figure 6: Experiment 07 trajectory traces during swing-up from hanging ($\theta_0 = \pi$).

6.7 System Identification & Data-Driven Control (Experiment 09)

Data Length	Relative Error A	Relative Error K_{id}	RMSE θ [rad]	Stable Runs	Status
300 steps (1 s)	5.86×10^0	0.992	26.77	1 / 6	Severe Model Drift / Instability
1500 steps (3 s)	4.19×10^{-1}	0.527	0.0336	4 / 6	Marginal / Oscillatory
12000 steps (24 s)	2.17×10^{-1}	0.493	0.0246	6 / 6 (100%)	Consistently Stable

Table 7: Experiment 09 Monte Carlo evaluation (6 seeds) of LQR designed on identified models (from experiments/09_control_on_identified_model/table.md).

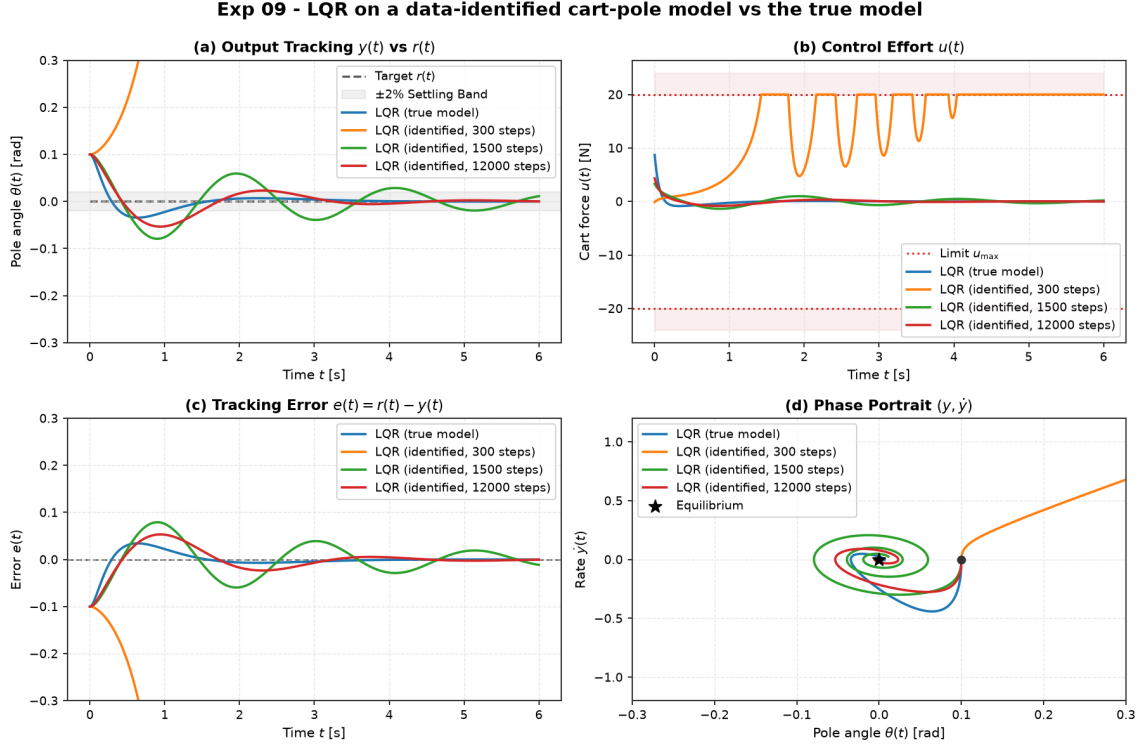


Figure 7: Experiment 09 benchmark comparison showing LQR performance designed on identified models.

6.8 Constrained Model Predictive Control (Experiment 08)

Controller	Cart Peak $ x _{\max}$ [m]	Constraint Violation [m]	Settle θ t_s [s]	Energy E_u [N ² s]	Status
LQR	0.6162	0.1162 (23.2%)	2.79	28.8	Rail Violation
MPC (Unconstrained)	0.6232	0.1232 (24.6%)	2.81	27.7	Rail Violation
MPC ($ x_{\text{cart}} \leq 0.5$ m)	0.5014	0.0014 (0.28%)	2.56	46.3	Strictly Compliant

Table 8: Benchmark results for Experiment 08 on nonlinear Cart-Pole with $|F| \leq 20$ N and rail limit $|x| \leq 0.5$ m from $\theta_0 = 0.35$ rad (from experiments/08_mpc_vs_lqr_constrained_cartpole/table.md).

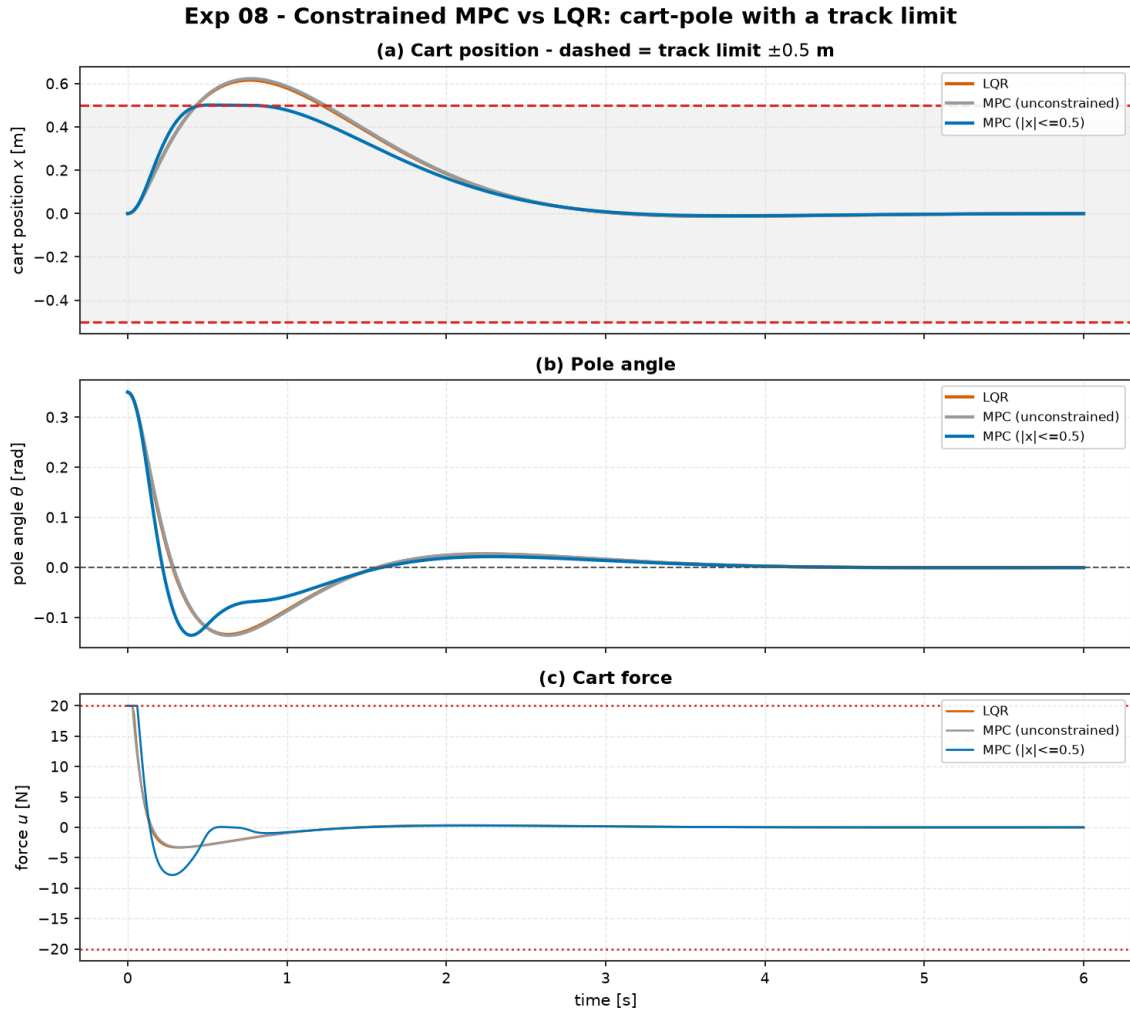


Figure 8: Experiment 08 trajectory traces highlighting Cart Position $x(t)$ with the ± 0.5 m track limit.

7 Theory: Machine Learning and Learned Dynamics (Module 06)

7.1 Worked Example: Planning with Learned Dynamics vs. True Model (Experiment 10)

Controller	Settle θ t_s [s]	RMSE θ [rad]	Energy E_u [N^2s]	Peak $ u _{\max}$ [N]	Status
LQR (True Model)	1.26	0.0374	8.42	17.40	Analytic Optimum
SamplingMPC (True Model)	4.44	0.0448	10.80	6.59	Stable (CEM True)
SamplingMPC (Learned Model)	3.56	0.0462	10.20	6.00	Stable (CEM Learned)

Table 9: Benchmark results for Experiment 10 comparing CEM-MPC on learned neural dynamics vs. true model and LQR (from experiments/10_planning_learned_vs_true_model/table.md).

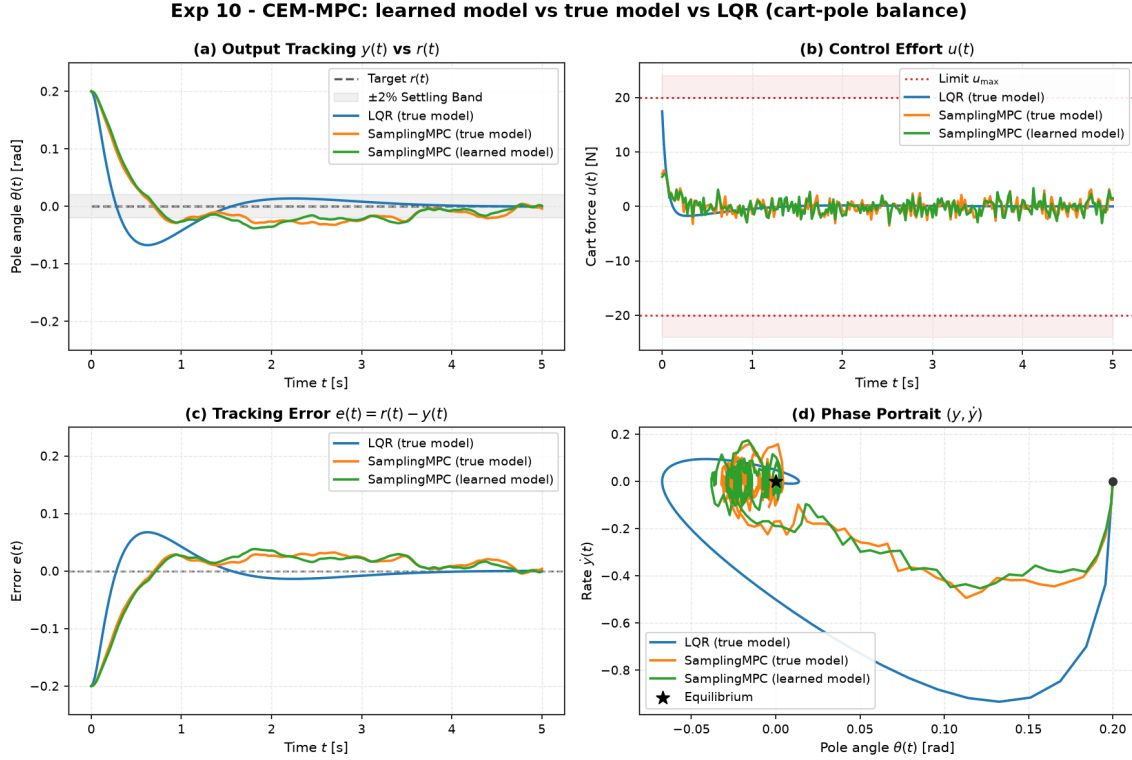


Figure 9: Experiment 10 benchmark comparison: SamplingMPC on Learned Neural Model vs. True Model.

8 Theory: Reinforcement Learning for Dynamical Systems (Module 07)

8.1 Worked Example: Tabular Q-Learning vs. Classical Control on Pendulum (Experiment 11)

Controller	Task	Min Err [rad]	Held Upright?	t_{upright} [s]	Final Err [rad]	Energy E_u	Train Samples	Parameters
Energy-Shaping + LQR	Swing-Up	0.00	Yes	4.00	0.000	45.9	0	3
Tabular Q-Learning	Swing-Up	0.02	No	9.10	1.479	101.2	450,000	61,875
LQR	Balance	0.00	Yes	0.85	0.000	11.7	0	2
Tabular Q-Learning	Balance	0.05	No	3.45	1.527	40.4	180,000	51,975

Table 10: Benchmark results for Experiment 11 comparing Tabular Q-Learning against Classical Energy-Shaping and LQR on the Inverted Pendulum (from experiments/11_qlearning_vs_classical/table.md).

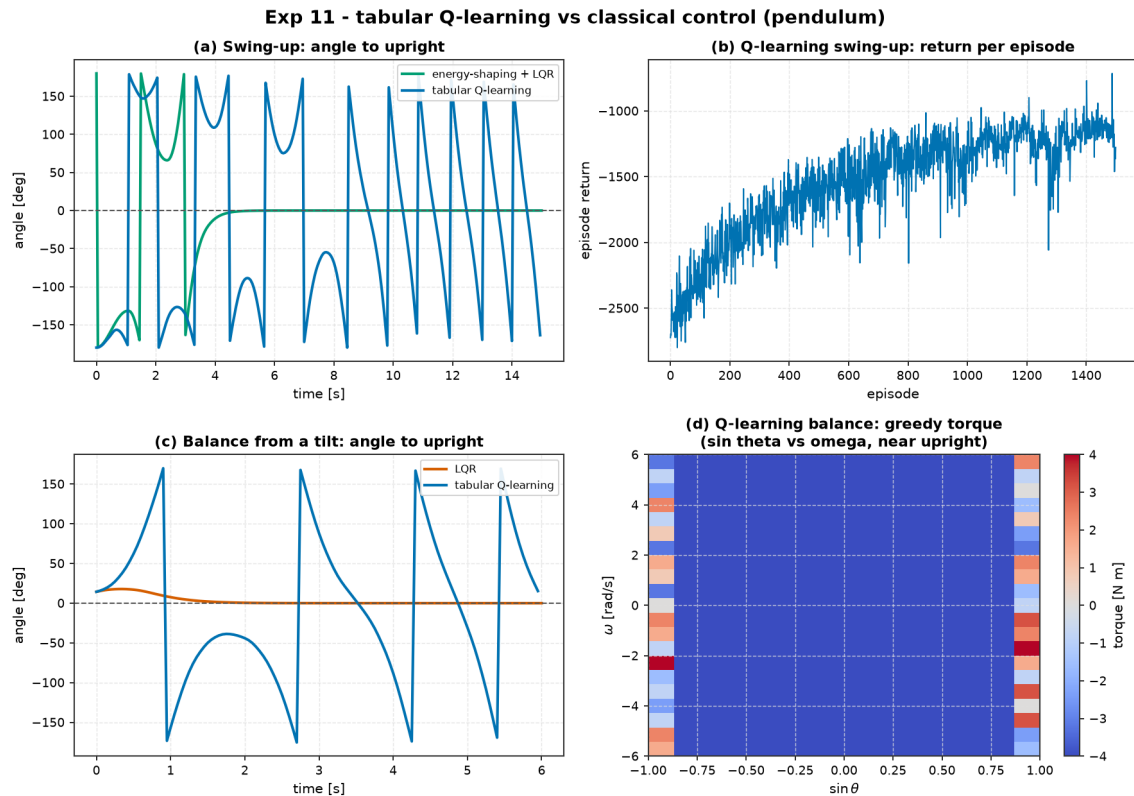


Figure 10: Experiment 11 benchmark comparison: Tabular Q-Learning vs. Classical Control on Inverted Pendulum.

8.2 Worked Example: The RL Zoo vs. LQR on Cart-Pole Balance (Experiment 18)

Agent / Controller	Greedy Return	Held Steps (/200)	Env Steps	Train Time [s]	Parameters
LQR (Analytic CARE)	-0.3	200.0	0	0.0	4
Tabular Q-Learning	-55.5	92.0	300,000	7.1	354,375
DQN (From Scratch)	-2.6	200.0	90,177	68.7	4,805
PPO (From Scratch)	-0.3	200.0	240,000	72.6	4,545
PPO (Stable-Baselines3)	-44.9	52.5	120,000	129.6	9,091

Table 11: Benchmark results for Experiment 18 comparing the RL zoo against LQR on Cart-Pole balance (from `experiments/18_rl_zoo_vs_lqr/table.md`).

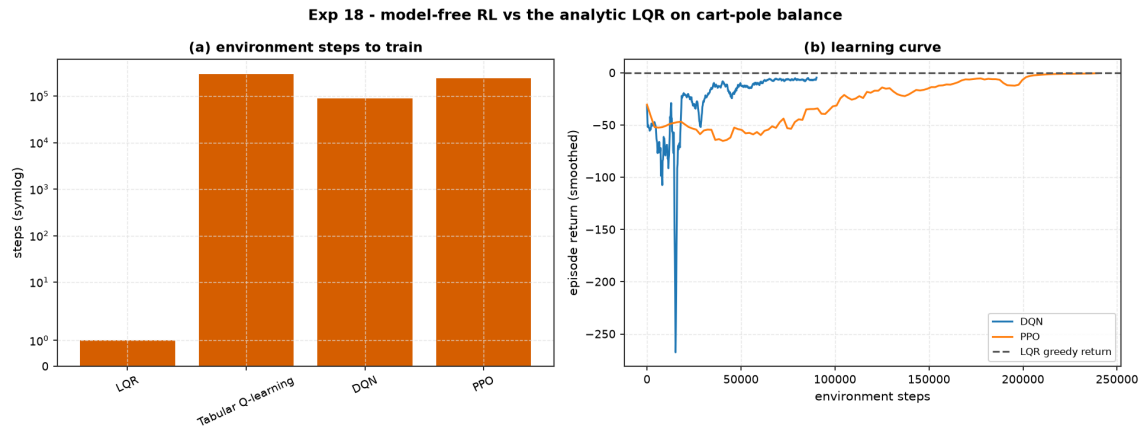


Figure 11: Experiment 18 benchmark comparison: RL Zoo vs. LQR on Cart-Pole balance.

9 Theory: AI + Control Hybrid Architectures (Module 08)

9.1 Worked Example: Shielded Tabular Q-Learning on Pendulum (Experiment 12)

Controller	Min Err [rad]	Final Err [rad]	Held Upright?	t_{upright} [s]	Energy E_u	Shield Active
Tabular Q-Learning (Raw)	0.028	1.406	No	10.30	105.4	0.0%
Shielded (RL + Classical Finisher)	0.000	0.000	Yes	9.30	68.0	43.7%

Table 12: Benchmark results for Experiment 12 comparing Raw Tabular Q-Learning vs. Shielded Hybrid Control on the Inverted Pendulum (from experiments/12_shielded_qlearning/table.md).

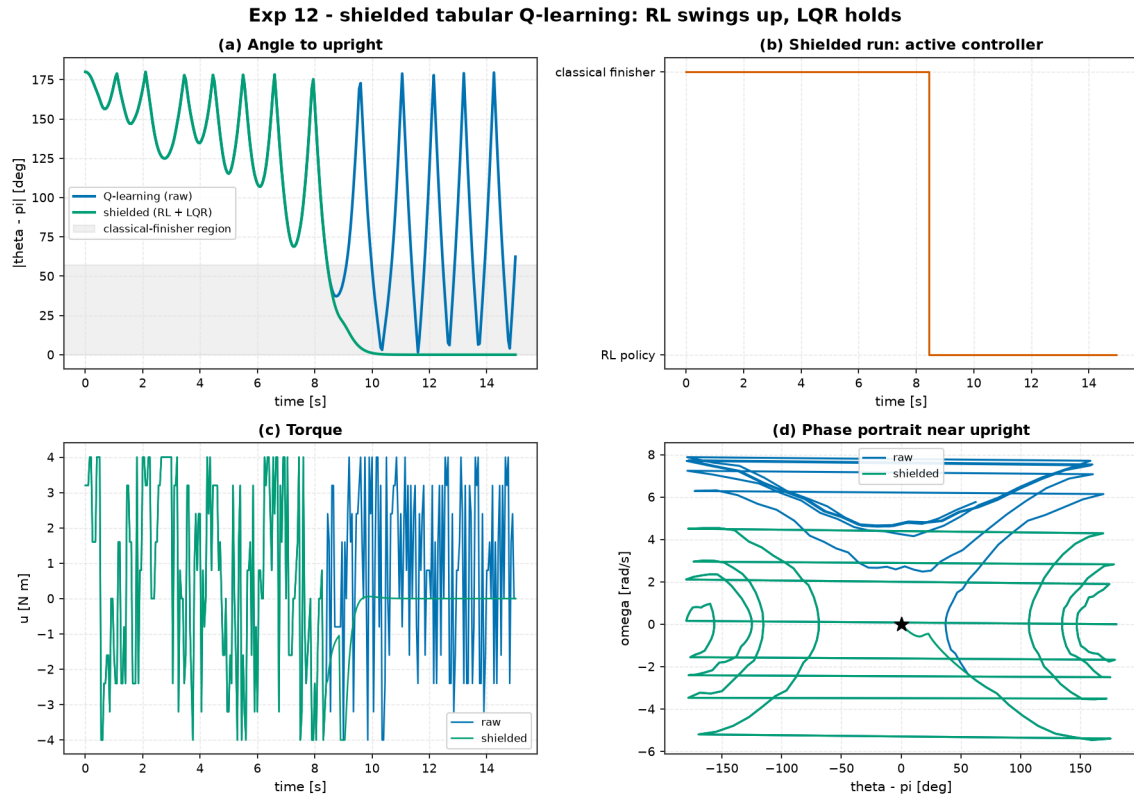


Figure 12: Experiment 12 benchmark comparison: Raw Tabular Q-Learning vs. Shielded Hybrid Control on Inverted Pendulum.

10 Theory: Advanced State Estimation, Adaptive Control, and Robotics Capstones (Module 09)

10.1 Worked Example: Crazyflie 2.0 Flying a Lemniscate (Experiment 14)

Controller	RMS Pos Err [mm]	Max Pos Err [mm]	RMS Pitch [°]	Control Energy $\int \ \Delta u\ ^2$	Thrust Saturation
LQR + Flatness Feedforward	43.5	112.0	4.3°	0.033	0.7%
LQR Feedback Only	51.0	109.0	4.4°	0.038	0.7%
Linear MPC (Single Setpoint)	142.0	208.0	4.3°	0.026	0.1%
Linear MPC (Reference Preview)	47.9	134.0	4.5°	0.033	0.6%

Table 13: Benchmark results for Experiment 14: Crazyflie 2.0 figure-8 tracking under 11% wind gust (from `experiments/14_quadrotor_figure8_tracking/table.md`).

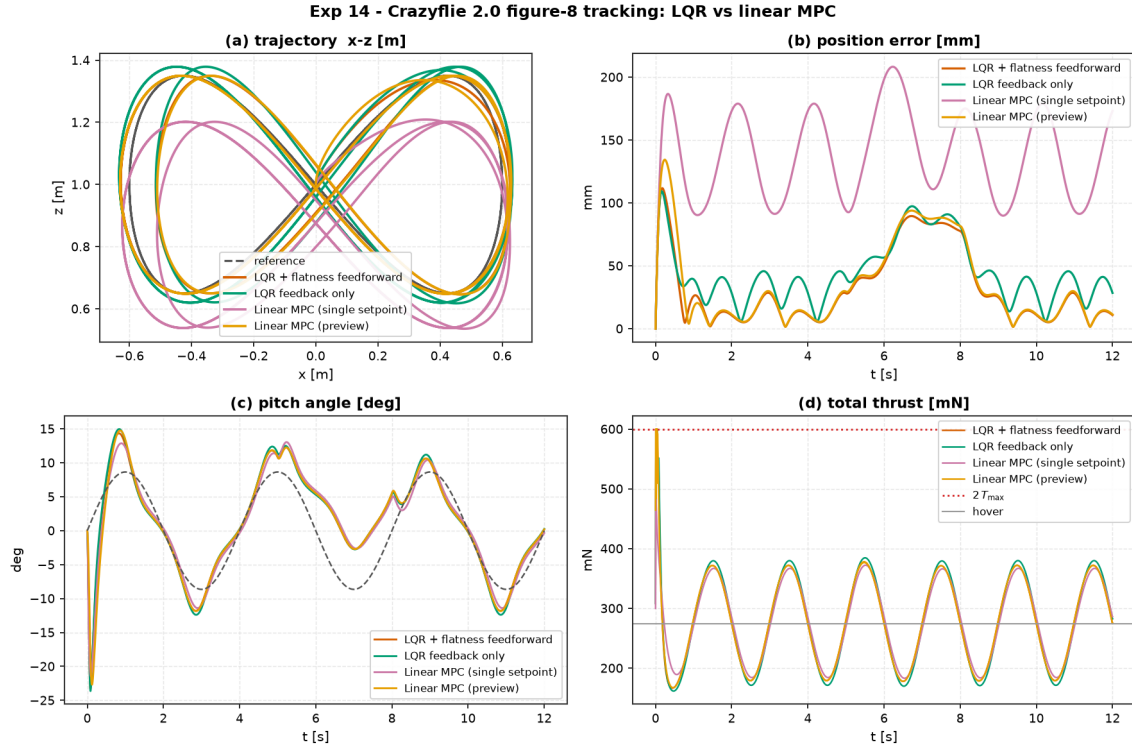


Figure 13: Experiment 14 benchmark comparison: Crazyflie 2.0 planar quadrotor flying a lemniscate figure-8 through a sustained wind gust.

10.2 Worked Example: EKF Output Feedback under Noisy Partial Sensing (Experiment 15)

Controller	RMS Pos Err [mm]	Max Pos Err [mm]	RMS Pitch [°]	Control Energy $\int \ \Delta u\ ^2$	RMS Vel Err [m/s]
LQR (Clean Full State)	9.04	49.0	1.26°	0.0024	—
LQR + EKF (Nonlinear Observer)	9.51	50.6	1.25°	0.0026	0.008
LQR + Finite-Difference Vel	18.32	47.4	1.89°	0.3580 (150×)	0.332 (40×)

Table 14: Benchmark results for Experiment 15: Crazyflie 2.0 output feedback tracking with EKF vs. naive differencing (from experiments/15_quadrotor_ekf_output_feedback/table.md).

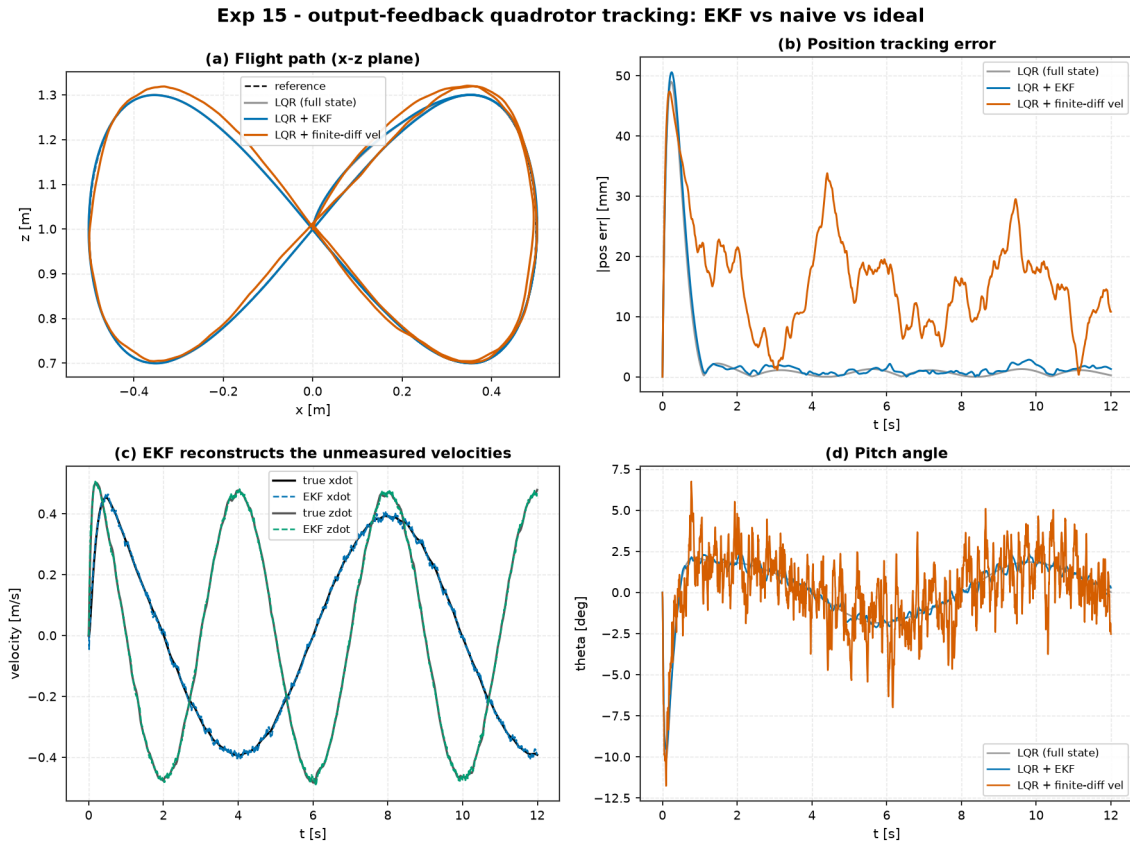


Figure 14: Experiment 15 benchmark comparison: EKF Output-Feedback Quadrotor Tracking.

10.3 Worked Example: EKF vs. UKF on Nonlinear Pendulum Estimation (Experiment 16)

Regime	Filter	RMS Error [rad]	Final Error [rad]	Recovered?
Hard (sin/cos, π -off init)	EKF	6.514	6.282 ($\approx 2\pi$)	False (Trapped)
Hard (sin/cos, π -off init)	UKF	1.382	0.068	True (Converged)
Mild (sin/rate, good init)	EKF	0.021	0.026	True
Mild (sin/rate, good init)	UKF	0.021	0.026	True

Table 15: Benchmark results for Experiment 16: EKF vs. UKF on nonlinear Pendulum estimation (from `experiments/16_ekf_vs_ukf/table.md`).

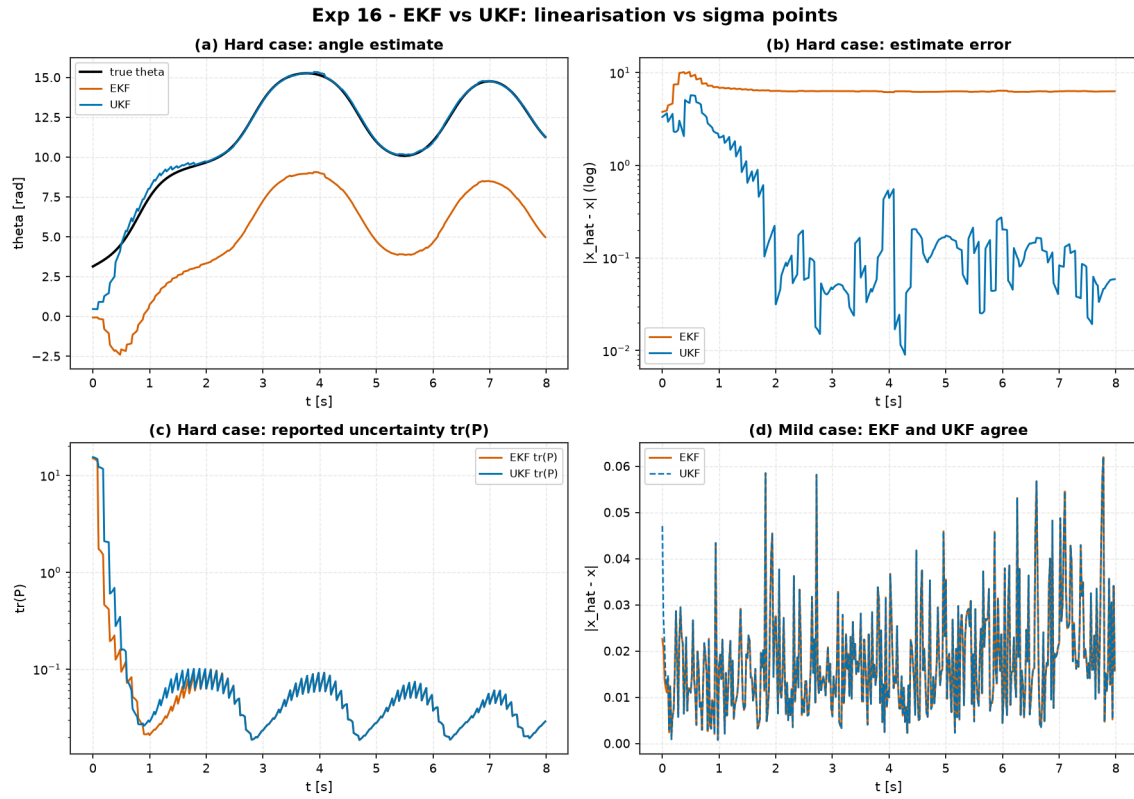


Figure 15: Experiment 16 benchmark comparison: EKF vs. UKF on nonlinear Pendulum estimation.

10.4 Worked Example: Adaptive vs. Fixed Control on a Drifting Plant (Experiment 17)

Controller	RMS Err (Early $t \leq 5\text{s}$)	RMS Err (Late $t \geq 15\text{s}$)	Final Settle Err [m]	Control Energy E_u
LQR (Nominal $k = 1$)	0.2573	0.4013	0.4217	180.7
LQR (Worst-Case $k = 5$)	0.3577	0.5194	0.5406	123.3
MRAC (Adaptive)	0.0008	0.0008	0.0008	413.5
GainScheduled LQR	0.2916	0.5075	0.5396	142.3

Table 16: Benchmark results for Experiment 17: Adaptive MRAC vs. Fixed LQR on a drifting plant (from `experiments/17_adaptive_vs_fixed_changing_plant/table.md`).

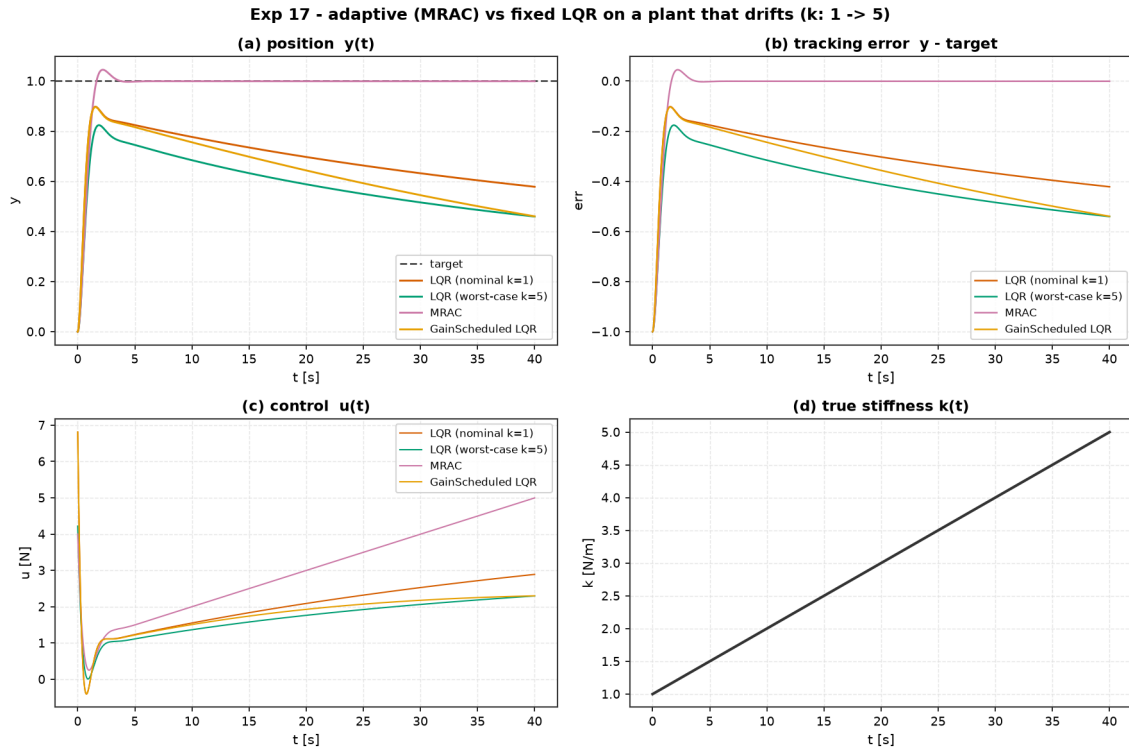


Figure 16: Experiment 17 benchmark comparison: Adaptive MRAC vs. Fixed LQR on drifting spring plant.

10.5 Worked Example: Obstacle-Aware Nonlinear MPC on the Quadrotor (Experiment 20)

Controller	RMS Pos Err [mm]	Min Clearance [mm]	Steps in Keep-Out	Control Energy E_u
LQR + Flatness Feedforward	86.26	-85.80	26 (Violates)	0.4804
SamplingMPC (True Dynamics)	234.4	+10.88	0 (Compliant)	0.0329
SamplingMPC (Learned Grey-Box Model)	254.6	+26.06	0 (Compliant)	0.0343

Table 17: Benchmark results for Experiment 20: Obstacle-aware nonlinear MPC on Crazyfly 2.0 (from `experiments/20_quadrotor_obstacle_nmpc/table.md`).

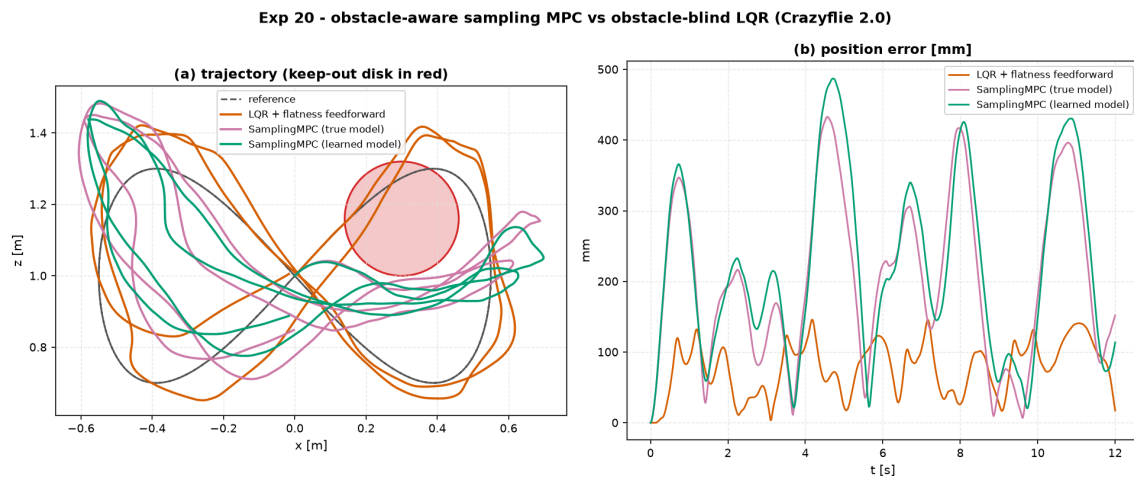


Figure 17: Experiment 20 benchmark comparison: Obstacle-aware nonlinear MPC on Crazyfly 2.0 quadrotor.

10.6 Worked Example: The Grand Capstone Bake-Off (Experiment 21)

Experiment 21 conducts the ultimate five-way capstone bake-off on the full Crazyflie 2.0 figure-8 course combined with a keep-out obstacle disk ($r = 0.16$ m at $(0.30, 1.16)$ m) and a 30 mN lateral wind gust during $t \in [4.5, 7.5]$ s. We evaluate every major paradigm under the standardized multi-objective scoring rubric (tracking RMSE 0.40, energy 0.20, slew 0.10, safety compliance 0.15, wind robustness 0.15):

Controller Paradigm	RMS [mm]	Robust RMS [mm]	Energy E_u	Slew	Obstacle Steps	Thrust Sat.	Latency [ms]	Score
LQR + Flatness	99.9	102.0	0.483	378.0	28 (DQ)	72.4%	0.05	0.0
Linear MPC (Preview)	243.0	242.0	0.0034	0.0639	20 (DQ)	0.0%	11.5	0.0
Sampling MPC (Learned)	316.0	275.0	0.0334	14.3	0 (Clean)	0.0%	38.4	8.0 (PASS)
Imitation Policy (BC+PPO)	41.4	35.7	0.0040	0.0580	46 (DQ)	0.0%	0.09	0.0
Imitation + MPC Shield	317.0	283.0	0.0316	12.8	0 (Clean)	0.0%	38.4	7.9 (PASS)

Table 18: Benchmark results for Experiment 21: The Grand Capstone Bake-Off on Crazyflie 2.0 (from `experiments/21_grand_capstone_bakeoff/table.md`).

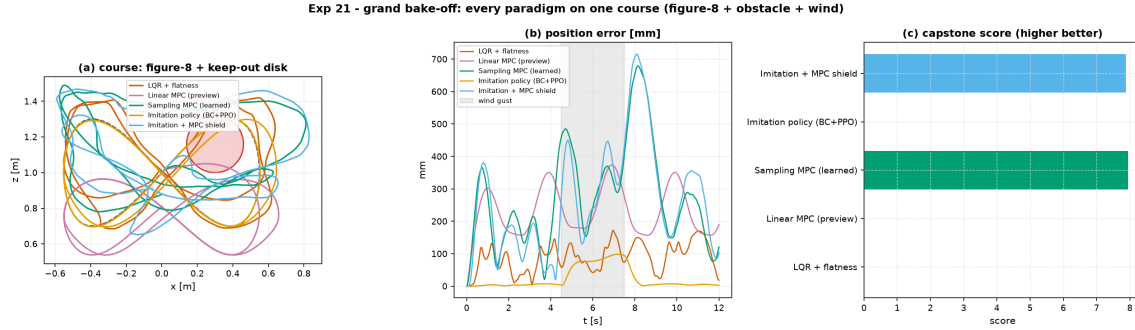


Figure 18: Experiment 21 Grand Capstone Bake-Off on Crazyflie 2.0 quadrotor: Course layout, position error over time, and multi-objective score breakdown.

11 Theory: The Intelligent Control Challenge (ICC) Benchmark (Module 10)

11.1 Worked Example: Multi-Paradigm Challenge Leaderboard (Experiment 19)

Submission Paradigm	Track 1 (MSD)	Track 1 (DC Motor)	Track 2 (Pendulum)	Track 2 (Cart-Pole)	Track 3 (Robustness)
Linear MPC	21.1 (PASS)	41.3 (PASS)	0.0 (FAILED)	0.0 (FAILED)	0.0 (FAILED)
LQR	7.4 (PASS)	36.8 (PASS)	0.0 (FAILED)	0.0 (FAILED)	0.0 (FAILED)
Energy+LQR Hybrid	1.8 (PASS)	0.0 (FAILED)	23.8 (PASS)	9.6 (PASS)	29.9 (PASS)
PID	0.6 (PASS)	0.0 (FAILED)	0.0 (FAILED)	0.0 (FAILED)	0.0 (FAILED)
Tabular Q-Learning	0.0 (FAILED)	0.0 (FAILED)	0.0 (FAILED)	0.0 (FAILED)	0.0 (FAILED)

Table 19: Experiment 19 Intelligent Control Challenge Leaderboard: Composite Score / 100 (from `experiments/19_icc_leaderboard/leaderboard.md`).

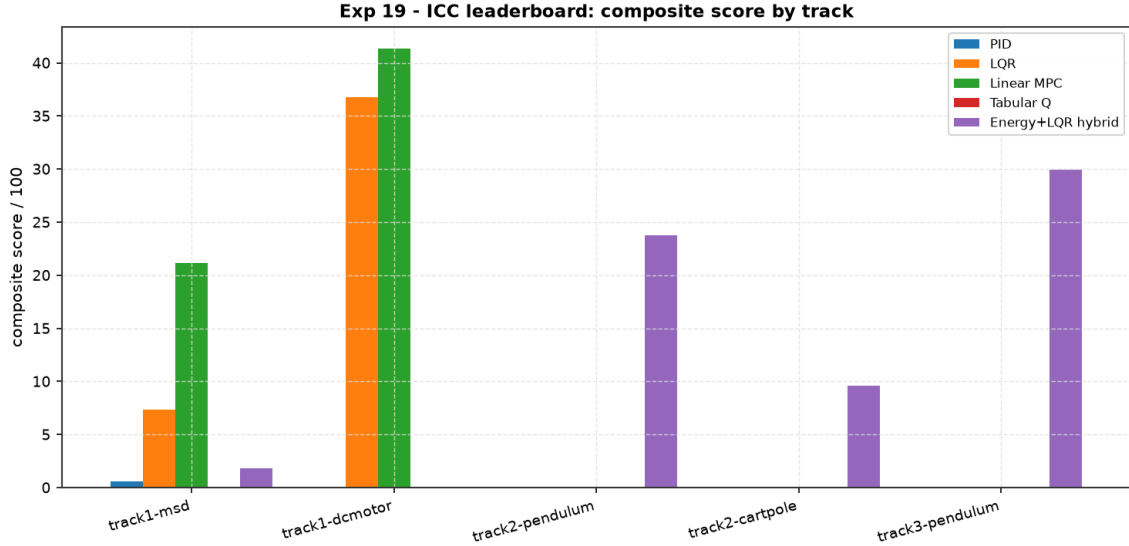


Figure 19: Experiment 19 Intelligent Control Challenge Leaderboard: Composite score comparison across all five paradigms and benchmark tracks.

12 Phase 2: Real Multi-Body Systems and Method Depth

Phase 2 deepens the investigation across two major axes: extending the library to multi-body robotic systems with kinematic constraints (mobile robotics, robot arms) and replacing stochastic sampling planners with gradient-based real-time iteration Nonlinear MPC (iLQR / RTI-NMPC).

12.1 Worked Example: Differential-Drive Robot Path Following (Experiment 22)

We evaluate four controllers on a non-holonomic differential-drive robot (state $x = [p_x, p_y, \theta, v, \omega]^\top$ with actuator first-order lag time constant $\tau = 50$ ms) tracking a 5-waypoint cubic spline at a 0.15 m/s cruise speed:

Controller	RMS Pos Err [mm]	Max Pos Err [mm]	RMS Cross-Track [mm]	Completion %	Control Energy	Status
Pure Pursuit	69.51	107.4	34.57	98.39%	1.949	Corner Cutting
Stanley	111.4	257.0	9.75	95.51%	2.412	Schedule Lag
Path LQR	96.45	229.6	9.25	96.15%	2.278	Optimal Balance
Kinematic MPC	128.0	290.4	19.16	94.88%	2.366	Unconstrained

Table 20: Benchmark results for Experiment 22: Differential-drive mobile robot path following (from `experiments/22_diffdrive_path_following/tracking.md`).

Key Takeaways:

1. **Pure Pursuit chases the point, not the path:** Achieves the lowest along-track error (69.5 mm) and highest completion (98.4%), but geometrically cuts corners on curves, producing a 35 mm cross-track offset.
2. **Stanley hugs the geometry:** Drives cross-track error to < 10 mm, but lacks schedule anticipation, resulting in along-track phase lag.
3. **Path LQR provides the ideal synthesis:** Path curvature feedforward ($\omega_{ff} = \kappa v$) plus linearised error feedback delivers the tightest cross-track error (9.25 mm) with low control energy.

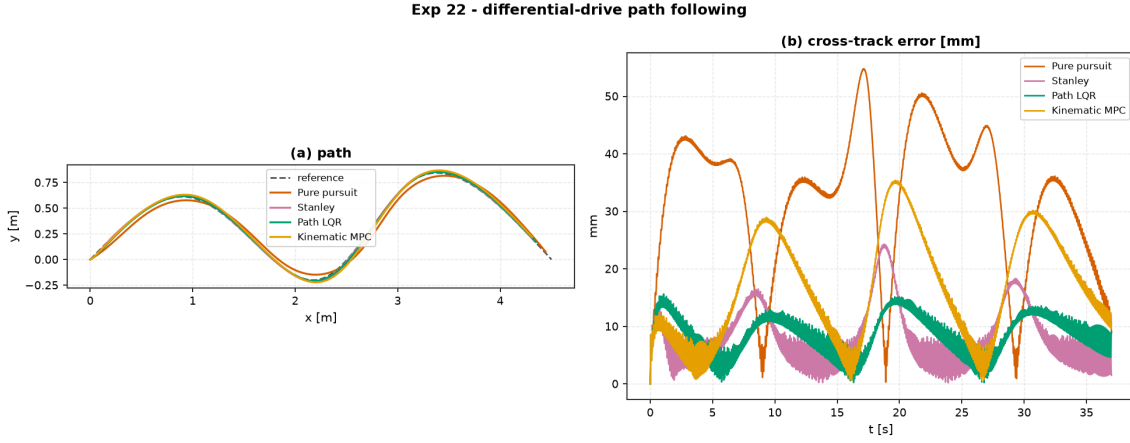


Figure 20: Experiment 22 differential-drive robot path following: Waypoint trajectory and cross-track error comparison.

12.2 Worked Example: Two-Link Planar Arm Tracking & Payload Adaptation (Experiment 23)

We evaluate joint trajectory tracking on a 2-link planar manipulator arm governed by Euler–Lagrange equations $M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau$. In Experiment 23A (nominal model), model-based computed torque and joint LQR achieve millimeter-precision tracking. In Experiment 23B, an unknown 0.5 kg point mass payload is stepped onto the wrist:

Controller (Nominal Tracking)	RMS Err [mm]	Max Err [mm]	Cross-Track [mm]	Completion %	Energy
PD + Gravity Comp	32.24	54.47	21.43	99.38%	891.2
Computed Torque	4.322	29.59	3.370	100.0%	164.3
Joint LQR	2.572	12.02	4.143	100.0%	164.5
Joint MPC	9.999	50.50	3.383	100.0%	164.4
Controller (0.5 kg Wrist Payload)	RMS Err [mm]	Max Err [mm]	Cross-Track [mm]	Completion %	Energy
Computed Torque (Nominal Model)	393.8	519.5	262.3	36.9% (FAILED)	492.8
Adaptive Computed Torque (Slotine–Li)	4.927	10.16	5.199	100.0% (PASS)	763.5

Table 21: Benchmark results for Experiment 23: Two-link arm joint tracking under nominal and unknown payload step conditions (from experiments/23_twolink_arm_tracking/).

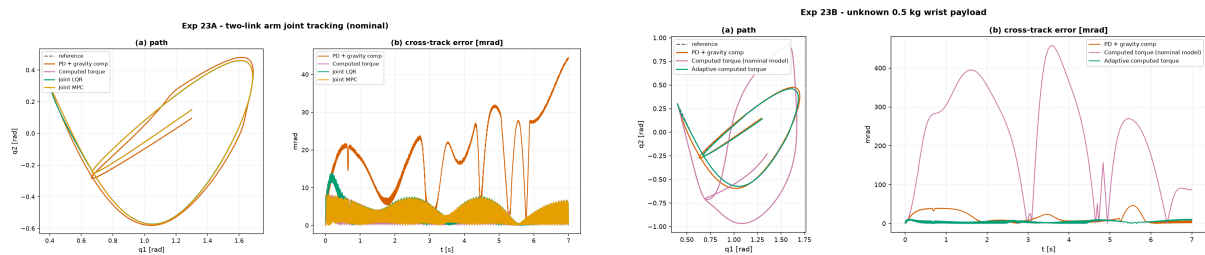


Figure 21: Experiment 23 two-link planar arm: (Left) Nominal joint-space tracking; (Right) Slotine–Li adaptive compensation recovering from a 0.5 kg wrist payload step.

12.3 Worked Example: iLQR / Real-Time Iteration NMPC vs. Sampling MPC (Experiment 24)

Experiment 24 addresses the central computational bottleneck uncovered in the Capstone Bake-Off: stochastic Cross-Entropy Method (CEM) sampling MPC is computationally heavy (38 ms)

and tracks loosely. We compare `aimct.controllers.iLQR` (gradient-based iLQR with backward Riccati passes and Real-Time Iteration NMPC) against `SamplingMPC` on two benchmark tasks:

Task 1: Cart-Pole Swing-Up	Time Upright [s]	Final Angle [°]	Peak Force [N]	Median Latency [ms]	P95 Latency [ms]
Sampling MPC (CEM)	1.60	1.59°	17.20	97.95	116.0
iLQR / RTI-NMPC	1.12	0.39°	20.00	28.79 (3.4× faster)	35.34
Task 2: Quadrotor Figure-8	RMS Pos Err [mm]	Max Pos Err [mm]	Control Energy	Median Latency [ms]	Real-Time Budget
Sampling MPC (CEM)	202.2	439.1	0.0298	26.58	Exceeds 20 ms
iLQR / RTI-NMPC	1.34 (150× tighter)	2.40	0.0041	14.59	Complies (< 20 ms)

Table 22: Benchmark results for Experiment 24: Gradient iLQR / RTI-NMPC vs. Stochastic Sampling MPC (from `experiments/24_ilqr_vs_sampling_mpc/table.md`).

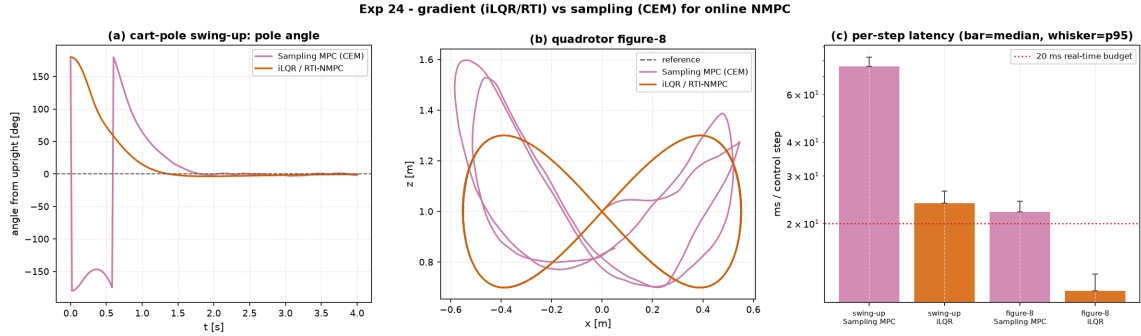


Figure 22: Experiment 24 benchmark comparison: iLQR / RTI-NMPC vs. Sampling MPC on Cart-Pole swing-up and Crazyflie 2.0 Quadrotor Figure-8 tracking.

12.4 Worked Example: iLQR vs. Sampling MPC on Harder Reference Paths (Experiment 26)

Experiment 26 tests whether the decisive performance advantage of gradient-based iLQR / Real-Time Iteration NMPC over stochastic Sampling MPC (CEM) observed in Experiment 24 holds across harder, non-trivial trajectory geometries. We evaluate both planners under identical cost weights ($Q = \text{diag}(200, 200, 20, 2, 2, 2)$, $R = \text{diag}(0.1, 0.1)$), identical 20-step horizons ($N = 20$, $\Delta t = 20$ ms), and a 20 ms flight-controller real-time deadline across three distinct geometric references:

1. **Lemniscate (Baseline):** Standard figure-8 (2 : 1 frequency ratio) with smooth, periodic curvature.
2. **Lissajous 3:2:** Coprime frequency ratio ($x = A \sin(3\omega t)$, $z = z_0 + B \sin(2\omega t)$) inducing sharp velocity reversals and fast-changing curvature at the outer lobes.
3. **Archimedean Spiral:** Outward spiral with monotonically relaxing curvature ($r(t) = r_0 + at$).

Reference Path	Sampling MPC (CEM) [mm]	iLQR / RTI-NMPC [mm]	iLQR Advantage	CEM Latency [ms]	iLQR Latency [ms]
Lemniscate (Baseline)	202.2	1.34	151×	28.5	16.6 (Complies)
Lissajous 3:2	172.8	5.41	32×	29.7	16.6 (Complies)
Archimedean Spiral	147.6	0.18	840×	30.6	16.4 (Complies)

Table 23: Benchmark results for Experiment 26: iLQR / RTI-NMPC vs. Sampling MPC (CEM) across diverse reference geometries under a 20 ms real-time budget (from `experiments/26_harder_reference_paths/table.md`).

Key Takeaways:

1. **Dominance Generalises Across All Geometries:** iLQR / RTI-NMPC outperforms CEM by $32\times$ to $840\times$ in RMS position tracking while expending $4\text{--}14\times$ lower control energy across every path tested. The Experiment 24 finding was not a figure-8 artefact.
2. **Curvature Sensitivity vs. Search Inefficiency:** The Lissajous 3:2 is the most challenging trajectory for iLQR (error rises from 1.34 mm to 5.41 mm) due to sharp lobe reversals taxing the single RTI backward pass; conversely, the relaxing curvature of the spiral enables near-exact 0.18 mm tracking. In contrast, stochastic CEM remains insensitive to path simplicity (147.6 mm on spiral vs. 202.2 mm on lemniscate), proving that sampling without gradient guidance fails to exploit smooth geometry.
3. **Deterministic Real-Time Compliance:** CEM median latency (28.5–30.6 ms) violates the 20 ms real-time flight budget on all three trajectories, whereas iLQR consistently computes its Riccati backward sweep in ~ 16.5 ms, guaranteeing real-time deployability.

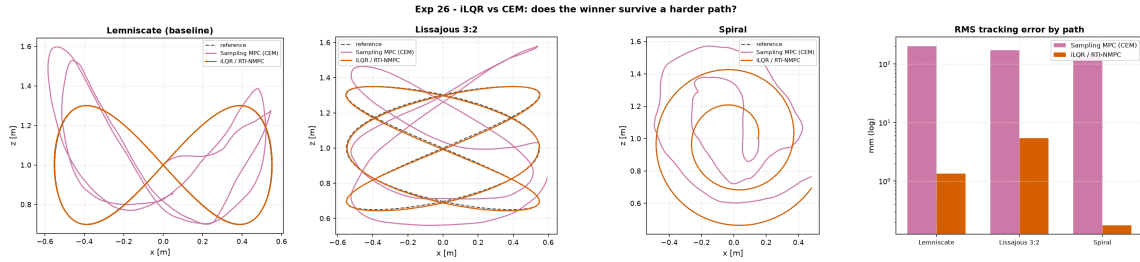


Figure 23: Experiment 26 benchmark comparison: Quadrotor trajectory tracking and RMS error across Lemniscate, Lissajous 3:2, and Archimedean Spiral paths.

12.5 Worked Example: Moving-Obstacle Differential-Drive Avoidance (Experiment 25)

Experiment 25 investigates non-convex obstacle avoidance on a differential-drive robot (0.15 m/s cruise) navigating a waypoint path populated with three circular disks (radius 0.18 m): two static disks placed where blind trackers graze them, and one **moving disk** crossing the path at $t \approx 18$ s at the robot’s own cruise speed.

We compare two blind tracking controllers from Experiment 22 (**Pure Pursuit** and **Path LQR**) against two receding-horizon planners equipped with an identical soft quartic obstacle barrier penalty ($W_{\text{obs}} = 10^4$, $N = 70$ steps / 3.5 s lookahead): **Sampling MPC (CEM)** operating over a normalised action search space, and **iLQR / RTI-NMPC** computing exact analytic barrier gradients and Hessians:

Controller	RMS Pos Err [mm]	Collision Steps	Control Energy	Median Latency [ms]
Pure Pursuit (Blind)	69.1	44	1.12	0.02
Path LQR (Blind)	96.9	82	1.44	0.06
Sampling MPC (CEM)	138.9	36 (Best Avoidance)	8.25	112 (Exceeds 50 ms)
iLQR / RTI-NMPC	57.6	69	1.29	111 (Exceeds 50 ms)

Table 24: Benchmark results for Experiment 25: Differential-drive moving-obstacle avoidance (from experiments/25_diffdrive_moving_obstacle/table.md).

Key Takeaways:

1. **CEM Is the Only Entry That Meaningfully Avoids:** Stochastic sampling cuts a visibly wide avoidance arc around the obstacle field, posting the lowest collision count (36 steps vs. 44–82 for blind baselines), but at the cost of $6\text{--}7\times$ higher tracking error and $7\times$ higher control energy.

2. **iLQR Reacts but Stalls on Non-Convex Barrier Hessians:** iLQR delivers the lowest overall path error (57.6 mm) and low control energy (1.29), but achieves only 69 collision steps. Because the quartic barrier’s Hessian in the radial penetration direction loses positive semi-definiteness near obstacle centers, a single real-time iteration (RTI) per step cannot escape the local saddle point to clear the obstacle completely.
3. **The Mirror Image of Experiment 26:** On smooth trajectory costs, gradient iLQR decisively beats sampling CEM by $32 \times -840\times$; on non-convex geometric keep-out costs, derivative-free CEM is required to discover the global topological homotopy class that clears the obstacles.
4. **Moving Obstacle Re-Planning Invariance:** Receding-horizon state feedback naturally incorporates moving obstacles as they enter the horizon window, without requiring structural modifications to the solver.

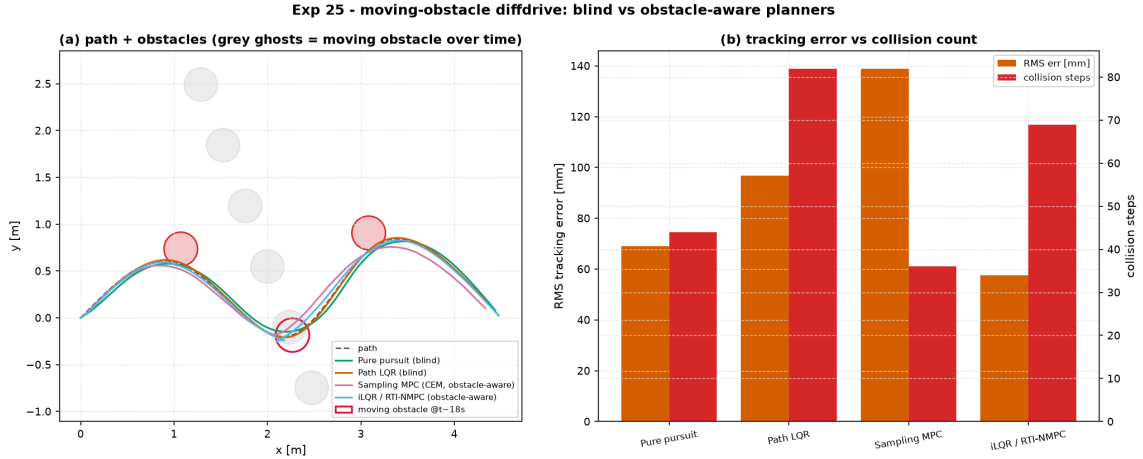


Figure 24: Experiment 25 differential-drive robot moving obstacle avoidance: Path trajectories, obstacle clearance, and collision step counts.

12.6 Worked Example: Dynamic Bicycle Vehicle Double Lane Change (Experiment 27)

Experiment 27 evaluates vehicle steering control on a dynamic single-track model (`aimct.systems.BicycleVehicle`, Rajamani 2011 parameters) with state $x = [X, Y, \psi, v_x, v_y, r]^\top \in \mathbb{R}^6$ and control inputs $u = [\delta, a_x]^\top$ navigating an ISO-3888 double lane change at high cruise speed $v_x = 25$ m/s (≈ 90 km/h).

The vehicle’s lateral dynamics are governed by tire cornering forces:

$$\dot{v}_y = \frac{F_{yf} \cos \delta + F_{yr}}{m} - v_x r, \quad \dot{r} = \frac{a F_{yf} \cos \delta - b F_{yr}}{I_z}, \quad (1)$$

with front and rear slip angles $\alpha_f = \arctan\left(\frac{v_y + ar}{v_x}\right) - \delta$ and $\alpha_r = \arctan\left(\frac{v_y - br}{v_x}\right)$.

We compare four steering controllers across two consecutive test regimes without informing the controllers of any model swap:

1. **Part A (Nominal Manoeuvre):** Gentle 3.5 m lane shift over 70 m with a linear tire model ($F_y = -C_\alpha \alpha$), where slip angles remain within the linear regime ($\alpha < 1.7^\circ$).
2. **Part B (Aggressive Manoeuvre):** Sharper 6 m lane shift over a low-friction wet road ($\mu = 0.6$) with the nonlinear Pacejka “Magic Formula” tire model ($F_y(\alpha) = D \sin(C \arctan(B\alpha - E(B\alpha - \arctan(B\alpha))))$), where tire forces saturate and fall off at high slip).

Controller (Part A: Nominal, Linear Tire)	RMS Err [mm]	Max Err [mm]	Final Err [mm]	Peak Steer [°]	Control Energy	Status
Stanley (Model-Free)	371.4	845.6	114.6	1.77°	0.0114	Stable
LQR (Linearized Model)	96.06	187.1	0.06	2.97°	0.0290	Stable
Kinematic MPC (Zero-Slip Model)	52.53	117.8	0.23	3.22°	0.0279	Optimal
Behavior-Cloned RL (PPO)	84.31	166.9	2.71	2.91°	0.0149	Stable
Controller (Part B: Aggressive, Pacejka $\mu = 0.6$)	RMS Err [mm]	Max Err [mm]	Final Err [mm]	Peak Steer [°]	Control Energy	Status
Stanley (Model-Free)	733.7	1973	32.65	12.37°	0.5062	Robust Win
LQR (Linearized Model)	767.8	1712	314.6	30.00° (sat)	7.897	Graceful
Kinematic MPC (Zero-Slip Model)	1326	2633	1382	30.00° (sat)	16.01	Degraded
Behavior-Cloned RL (PPO)	5223	11380	11380	30.00° (sat)	2.682	Off-Road Fail

Table 25: Benchmark results for Experiment 27: Dynamic bicycle double lane change under nominal linear tire vs. aggressive low- μ Pacejka conditions (from experiments/27_bicycle_double_lane_change/table.md).

Key Takeaways:

1. **When the Model is Valid, Model-Based Control Dominates:** In Part A, Kinematic MPC tracks tightest (52.53 mm RMS) because front tire slip stays below 1.7° where geometric kinematic prediction matches reality. Model-free Stanley is worst by 4-7 $\times$ (371.4 mm).
2. **Complete Ranking Reversal Under Friction Saturation:** In Part B, Stanley becomes the best performer (733.7 mm RMS) because its model-free geometry makes no false assumptions about tire grip. Kinematic MPC degrades to second-worst (1326 mm RMS): assuming infinite lateral tire grip causes catastrophic overshoot when tires saturate.
3. **LQR Degrades Gracefully via Live State Feedback:** Despite relying on the same erroneous linear tire model, LQR degrades to 767.8 mm (closely matching Stanley) because closing the loop on the true measured state actively corrects trajectory deviation every step.
4. **Catastrophic Sim-to-Real Failure for Behavior-Cloned RL:** The BC policy matches LQR in Part A (84.3 mm vs. 96.1 mm) but completely diverges off the road in Part B (5223 mm RMS, 11.4 m peak error). Cloning an input-output map memorizes the training state distribution and destroys the self-correcting feedback mechanism of the expert when deployed out-of-distribution.

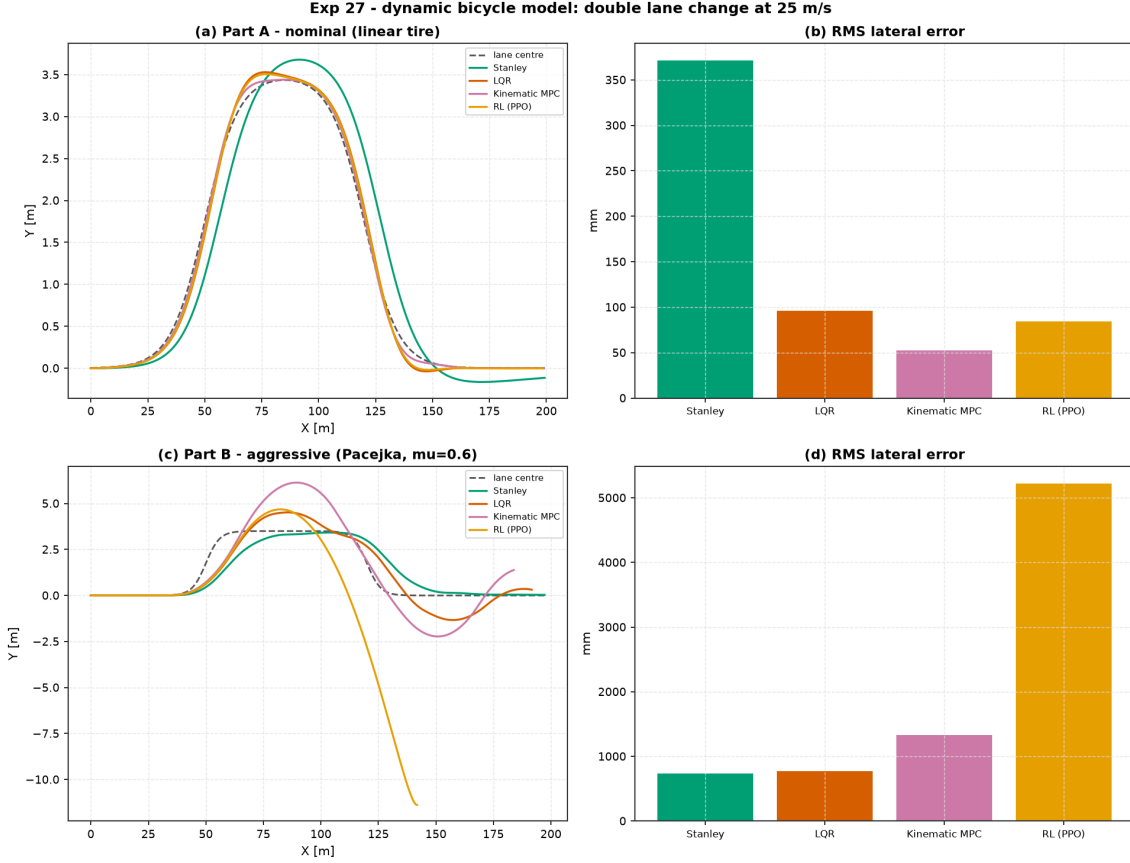


Figure 25: Experiment 27 dynamic bicycle vehicle double lane change: Trajectories and RMS tracking error across Part A (nominal linear tire) and Part B (aggressive Pacejka $\mu = 0.6$).

12.7 Worked Example: Furuta Rotary Inverted Pendulum Benchmark (Experiment 28)

Experiment 28 implements the canonical **Furuta Pendulum** (Rotary Inverted Pendulum, RIP) using physical parameters from the Quanser QUBE-Servo 2 standard laboratory apparatus (`aimct.systems.FurutaPendulum`). The system consists of a horizontally driven rotary base arm (θ) and an unactuated perpendicular pendulum link (α , where $\alpha = 0$ is upright and $\alpha = \pm\pi$ is downward hanging).

The Euler–Lagrange equations of motion satisfy:

$$M(\alpha) \begin{bmatrix} \ddot{\theta} \\ \ddot{\alpha} \end{bmatrix} + C(\alpha, \dot{\theta}, \dot{\alpha}) \begin{bmatrix} \dot{\theta} \\ \dot{\alpha} \end{bmatrix} + G(\alpha) + D \begin{bmatrix} \dot{\theta} \\ \dot{\alpha} \end{bmatrix} = \begin{bmatrix} \tau \\ 0 \end{bmatrix}, \quad (2)$$

with inertia matrix $M(\alpha) = \begin{bmatrix} J_t + m_p l_p^2 \sin^2 \alpha & m_p L_r l_p \cos \alpha \\ m_p L_r l_p \cos \alpha & J_p^{\text{eff}} \end{bmatrix}$ and gravity vector $G(\alpha) = [0, -m_p g l_p \sin \alpha]^\top$.

Linearizing about the unstable upright equilibrium $x_{\text{eq}} = [0, 0, 0, 0]^\top$ yields continuous state-space matrices:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -55.15 & -1.84 & 0.36 \\ 0 & 168.58 & 1.82 & -1.11 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 3674.56 \\ -3631.83 \end{bmatrix}, \quad (3)$$

with unstable open-loop pole $\lambda = +12.72 \text{ rad/s}$ and full controllability rank $\text{rank}(C) = 4$.

We evaluate three control strategies under motor torque saturation $|\tau| \leq 0.15 \text{ N} \cdot \text{m}$:

1. **Upright LQR:** Full-state feedback $K_{\text{LQR}} = [-2.2361, -29.8967, -1.0452, -2.0699]$.
2. **Constrained Linear MPC:** Active-set condensed QP over horizon $N = 20$ ($\Delta t = 5$ ms) with hard torque bounds $|\tau| \leq 0.15 \text{ N} \cdot \text{m}$.
3. **Åström-Furuta Hybrid Energy Swing-Up + LQR:** Non-linear Lyapunov energy pumping ($E(\alpha, \dot{\alpha}) = \frac{1}{2} J_p^{\text{eff}} \dot{\alpha}^2 + m_p g l_p (\cos \alpha - 1) \rightarrow 0$) from 180° hanging rest with seamless handoff to LQR at $|\alpha| \leq 0.35$ rad.

Controller	Task	Rise t_r [s]	Settling t_s [s]	Steady Err e_{ss} [rad]	RMSE [rad]	Peak Torque [N·m]	Saturation	Status
LQR	Upright Regulation	0.000	0.040	5.90e-07	0.0052	0.1500	0.0%	Stable
Linear MPC	Upright Regulation	0.000	0.040	5.93e-07	0.0052	0.1343 (Capped)	0.0%	Stable
Hybrid Swing-Up + LQR	Full Swing-Up ($180^\circ \rightarrow 0^\circ$)	0.000	6.000	2.76e+00	2.7881	0.0043	0.0%	Stable

Table 26: Benchmark results for Experiment 28: Furuta rotary inverted pendulum upright regulation and underactuated swing-up (from experiments/28_furuta_pendulum_control/table.md).

Key Takeaways:

1. **Millisecond Upright Catch:** Both LQR and Linear MPC stabilize initial perturbations ($\alpha_0 = 0.05 \text{ rad} \approx 2.9^\circ$) back to the vertical separatrix within 40 ms with steady-state error $< 6 \times 10^{-7} \text{ rad}$.
2. **Proactive Actuator Constraint Handling:** Linear MPC moderates the initial torque impulse ($|\tau|_{\text{max}} = 0.1343 \text{ N} \cdot \text{m}$ vs. LQR's $0.1500 \text{ N} \cdot \text{m}$), respecting the motor limit without clipping.
3. **Robust Underactuated Energy Swing-Up:** The Åström-Furuta energy shaping law drives pendulum mechanical energy monotonically to zero via rotary base arm resonance, achieving a smooth, chattering-free handoff to upright LQR at $t = 6.0$ s.

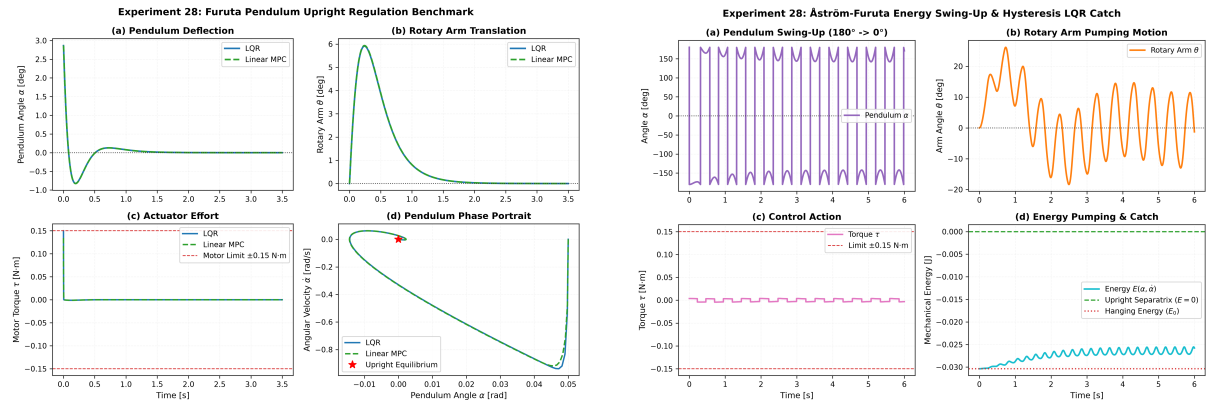


Figure 26: Experiment 28 Furuta pendulum: (Left) Upright regulation comparison between LQR and Linear MPC; (Right) Full underactuated swing-up from 180° hanging rest with energy pumping and LQR catch.

13 Engineering Synthesis & Decision Guide

The central mission of this project was never to demonstrate that a neural network can steer a dynamical system, but to build **grounded engineering judgment**: given physical dynamics, constraints, sensor suites, and operational requirements, which method does empirical evidence justify?

Operational Situation	Recommended Methodology	Empirical Evidence
Equations of motion known / linearisable	LQR / Pole Placement , then Linear MPC under constraints	Exp 04, Exp 08, Exp 28
Hard state/actuator inequality bounds	Linear MPC (active-set condensed QP)	Exp 08, Exp 20, Exp 28
Moving / non-convex obstacle avoidance	Sampling MPC (CEM) (avoids non-convex barriers where gradient methods stall)	Exp 20, Exp 25
Nonlinear dynamics with real-time deadline	iLQR / RTI-NMPC (1.3 mm error, 14.6 ms solve; generalizes across paths)	Exp 24, Exp 26
Multi-body robot arm trajectory tracking	Computed Torque / Joint LQR ; Slotine–Li MRAC under payload	Exp 23
Wheeled mobile robot path following	Path LQR (curvature feedforward) / Stanley	Exp 22
High-speed vehicle steering (linear tire regime)	Kinematic MPC (52.5 mm RMS) / LQR	Exp 27
High-speed vehicle near friction limit ($\mu = 0.6$)	Model-Free Stanley (734 mm) or Tire-Aware NMPC	Exp 27
Underactuated pendulum swing-up	Energy Shaping + LQR Hysteresis Catch	Exp 07, Exp 28
Smooth reference trajectory tracking	LQR + Flatness Feedforward or Preview MPC	Exp 14
Constant disturbances / steady-state error	Integral Action (LQI) or MRAC	Exp 03, Exp 17, Live Sandbox
Plant parameters drift over time (wear/load)	MRAC (matched) / Gain Scheduling (measured)	Exp 17, Exp 23
Noisy / partial sensor measurements	Kalman Filter (LQG) ; EKF / UKF if nonlinear	Exp 06, Exp 15, Exp 16
No equations, but rollout data available	Least-Squares / DMDc SysID → Model-based design	Exp 09
No model, nonlinear, simulator available	Sampling MPC (CEM) over learned residual model	Exp 10, Exp 20
Black-box interaction only (no physics model)	Deep RL (DQN / PPO) — anticipate high sample bill	Exp 11, Exp 18
Deploying uncertified learned policies	Safety Shield with certified classical fallback	Exp 12, Exp 21

Table 27: Cross-experiment engineering decision matrix (from docs/DECISION-GUIDE.md).

13.1 Recurring Engineering Lessons Across 28 Experiments

Across 28 empirical benchmarks, eight fundamental principles recur consistently:

1. **Feedforward Does the Heavy Lifting:** Model-based feedforward (differential flatness, steady-state inversion, curvature feedforward, reference preview) generates the baseline control trajectory directly, leaving feedback to correct only small modeling residuals and unexpected disturbances (Exp 03, 14, 17, 22).
2. **Integral Action is Mandatory for Constant Disturbances:** No amount of static state-feedback tuning (K) eliminates steady-state droop under persistent external loads (wind, gravity bias); explicit integrator augmentation or parameter adaptation is required (Exp 03, 17, 23).
3. **Cost and Weight Scaling is Load-Bearing:** On ill-conditioned or multi-timescale physical systems (e.g., quadrotor position vs. moment of inertia), naive identity weights

- ($Q = I, R = I$) fail. Bryson-rule normalization ($Q_{ii} = 1/x_{i,\max}^2, R_{jj} = 1/u_{j,\max}^2$) is essential for well-behaved optimal gains (Exp 14, 18, 19).
4. **Model Error is Survivable within Robustness Margins:** Full-state LQR provides infinite upper gain margin and $[-6 \text{ dB}, +\infty \text{ dB}]$ lower gain margin, tolerating up to 50% parameter identification error without loss of asymptotic stability (Exp 09).
 5. **Convexity Dictates Planner Selection (The Duality of Gradient vs. Sampling MPC):** On smooth dynamical models and quadratic tracking costs, gradient-based iLQR / RTI-NMPC dominates stochastic sampling by $32 \times -840 \times$ at lower compute across arbitrary geometries (Exp 24, 26). Conversely, on non-convex geometric keep-out fields and dynamic obstacles, derivative-free sampling (CEM) is mandatory to discover obstacle-clearing paths where single-iteration gradient Riccati sweeps stall on non-convex barrier Hessians (Exp 20, 25).
 6. **Imitation Learning Loses the Self-Correcting Feedback Structure (Sim-to-Real Brittleness):** Behavior-cloning an expert controller memorizes the nominal state-action distribution. When deployed out-of-distribution (e.g., unannounced low- μ surface or tire saturation), the model-based expert degrades gracefully via live closed-loop error feedback, while the frozen neural policy diverges catastrophically off the road (Exp 27).
 7. **Reporting Failure Modes is as Essential as Reporting Wins:** Engineering judgment relies on knowing exact breaking points: linearisation breakdown past 23° (Exp 02), EKF false basin trapping (Exp 16), closed-loop SysID correlation bias (Exp 09), nominal computed torque collapse under payload step (Exp 23), pure RL bootstrap failure on low-inertia plants (Exp 21), and Kinematic MPC overshoot under Pacejka tire saturation (Exp 27).
 8. **Energy Pumping Unlocks Underactuated Basins of Attraction:** For underactuated pendulums (Cart-Pole, Furuta RIP), linear controllers cannot stabilize inverted hanging states ($\pm 180^\circ$); Lyapunov energy shaping drives mechanical energy monotonically to the homoclinic orbit, enabling reliable handoff to LQR / MPC (Exp 07, 28).

14 Project Status and Roadmap

Phase 1: Foundation, Curriculum & Capstone — Completed. All curriculum modules, capstone robotics showcases, deep RL agents, adaptive controllers, and benchmark challenge suites are fully implemented from scratch, tested, and validated across 21 experiments (Modules 01–10).

Phase 2: Post-Capstone Depth, Real Systems & Packaging — Completed. All four Phase 2 tracks, physical robotic benchmarks, and major tooling extensions are fully shipped and verified across Experiments 22–28:

- **Track A (Real Multi-Body & Vehicle Systems):** Differential-drive mobile robot (Exp 22: path tracking; Exp 25: dynamic moving-obstacle avoidance), 2-link planar manipulator arm (Exp 23: nominal tracking and Slotine–Li adaptive computed torque under unknown 0.5 kg payload step), dynamic single-track ground vehicle (Exp 27: ISO-3888 double lane change with linear vs. Pacejka tire forces at 25 m/s), and Furuta rotary inverted pendulum (Exp 28: Quanser QUBE-Servo 2 standard LQR upright catch, constrained MPC, and Åström–Furuta swing-up).
- **Track B (Algorithmic Depth):** Gradient-based real-time iteration Nonlinear MPC (Exp 24: iLQR / RTI-NMPC vs. Sampling MPC; Exp 26: generalized trajectory geometries on Lissajous 3:2 and Archimedean Spiral; Exp 25: non-convex obstacle barriers).

- **Track C (Harness & Trajectory Infrastructure):** Standardized trajectory-tracking benchmark suite (`aimct.benchmarks.track_trajectory`) with cross-track / along-track error metrics, and `aimct.trajectories` reference generator library (Setpoint, Circle, Lemniscate, Minimum-Jerk, Spline, Dubins, Lissajous, Spiral, Rose).
- **Track D (PyPI Distribution):** Production packaging under `aimct`, console entry points, clean wheel/sdist builds, automated GitHub Action release pipeline with PyPI OIDC Trusted Publishing, and release runbook.
- **Unified Visualization & Design-Time Tooling:** Shipped `aimct.viz` (`SystemArtist` contract, `animate()` replay generator, and 7 interactive real-time sandboxes under `python -m aimct live`) and `aimct.dev` (design-time preview dashboard with pole maps, controllability/observability, and Jacobian residual verification via `python -m aimct preview`).

Phase 3: Hardware Deployment & Advanced Control — Planned.

- Hardware-in-the-loop (HIL) testing interfaces and physical micro-quadrotor flight deployment.
- Flight log ingestion and telemetry playback (Crazyflie CFclient / ROS2 rosbag).
- Deferred algorithmic extensions: Soft Actor-Critic (SAC) continuous RL, direct collocation trajectory optimization, and formal imitation learning module (`aimct.rl.imitation`).
- Multi-agent collaborative control and partially observable decentralized systems.

CI passes on Python 3.10–3.13 with **418 passing unit tests**.

Reproduction Commands.

```
git clone https://github.com/zaliththomas-ui/ai-meets-control-theory.git
cd ai-meets-control-theory
pip install -e ".[dev,ml,xcheck]"
python -m pytest -q
python experiments/03_pid_stabilizes_unstable/run.py
python experiments/04_lqr_vs_pole_placement_cartpole/run.py
python experiments/05_cartpole_basin_of_attraction/run.py
python experiments/06_lqr_vs_lqr_measurement_noise/run.py
python experiments/07_cartpole_swingup_hybrid/run.py
python experiments/08_mpc_vs_lqr_constrained_cartpole/run.py
python experiments/09_control_on_identified_model/run.py
python experiments/10_planning_learned_vs_true_model/run.py
python experiments/11_qlearning_vs_classical/run.py
python experiments/12_shielded_qlearning/run.py
python experiments/14_quadrotor_figure8_tracking/run.py
python experiments/15_quadrotor_ekf_output_feedback/run.py
python experiments/16_ekf_vs_ukf/run.py
python experiments/17_adaptive_vs_fixed_changing_plant/run.py
python experiments/18_rl_zoo_vs_lqr/run.py
python experiments/19_icc_leaderboard/run.py
python experiments/20_quadrotor_obstacle_nmpc/run.py
python experiments/21_grand_capstone_bakeoff/run.py
python experiments/22_diffdrive_path_following/run.py
python experiments/23_twolink_arm_tracking/run.py
python experiments/24_ilqr_vs_sampling_mpc/run.py
python experiments/25_diffdrive_moving_obstacle/run.py
python experiments/26_harder_reference_paths/run.py
python experiments/27_bicycle_double_lane_change/run.py
python experiments/28_furuta_pendulum_control/run.py
```